

# Exercises and Solutions

## Theory of Distributions

Prepared by ...

July 29, 2026

### Problem

Let  $X$  be a non-compact subset of  $\mathbb{R}^2$ . Prove that there is a continuous unbounded function from  $X$  to  $\mathbb{R}$ .

### Background

Recall that by the Heine–Borel theorem, a subset  $K \subseteq \mathbb{R}^2$  is compact if and only if it is closed and bounded. Thus, if  $X$  is non-compact, either

1.  $X$  is unbounded, or
2.  $X$  is bounded but not closed.

A function  $f : X \rightarrow \mathbb{R}$  is unbounded if for all  $M > 0$ , there exists  $x \in X$  such that  $|f(x)| > M$ .

### Solution

**Case 1.** Suppose  $X$  is unbounded. Then there exists a sequence  $\{x_n\} \subset X$  with  $\|x_n\| \rightarrow \infty$ . Define

$$f(x) = \|x\| = \sqrt{x_1^2 + x_2^2}.$$

This function is continuous on  $\mathbb{R}^2$  and unbounded on  $X$ .

**Case 2.** Suppose  $X$  is bounded but not closed. Then there exists a boundary point  $p \in \overline{X} \setminus X$ . Define

$$f(x) = \frac{1}{\|x - p\|}.$$

This function is continuous on  $X$  since  $p \notin X$ . Because there exists a sequence  $x_n \in X$  with  $x_n \rightarrow p$ , we have  $f(x_n) \rightarrow \infty$ . Hence  $f$  is unbounded.

### Conclusion

In both cases, there exists a continuous unbounded function  $f : X \rightarrow \mathbb{R}$ . Thus every non-compact subset of  $\mathbb{R}^2$  admits a continuous unbounded function.  $\square$

### Examples

1.  $X = \{(x, y) \in \mathbb{R}^2 : y > 0\}$  (unbounded). Function:  $f(x, y) = \sqrt{x^2 + y^2}$ .

2.  $X = \{(x, y) : x^2 + y^2 < 1\}$  (open unit disk). Function:  $f(x, y) = \frac{1}{1-(x^2+y^2)}$ .

## Problem

Consider the metric space  $(\mathbb{Q}, d)$  where  $\mathbb{Q}$  is the set of rational numbers and  $d$  is the Euclidean metric. Let

$$P = \{p \in \mathbb{Q} : 2 < p^2 < 3\}.$$

Prove that  $P$  is closed and bounded in  $\mathbb{Q}$ , but not compact. Give an open cover of  $P$  with no finite subcover.

## Solution

**Step 1. Boundedness.** If  $p \in P$ , then  $2 < p^2 < 3$ . Hence

$$\sqrt{2} < |p| < \sqrt{3},$$

so  $P \subset (-\sqrt{3}, -\sqrt{2}) \cup (\sqrt{2}, \sqrt{3})$ . Thus  $P$  is bounded in  $\mathbb{Q}$ .

**Step 2. Closedness.** The boundary of  $P$  in  $\mathbb{R}$  is given by the points  $\pm\sqrt{2}, \pm\sqrt{3}$ , all of which are irrational. Since these points are not in  $\mathbb{Q}$ ,  $P$  contains all its limit points in  $\mathbb{Q}$ . Hence  $P$  is closed in  $\mathbb{Q}$ .

**Step 3. Non-compactness.** Take a sequence  $p_n \in P$  with  $p_n \rightarrow \sqrt{2}$ . Since  $\sqrt{2} \notin \mathbb{Q}$ , this limit does not belong to  $P$ . Therefore  $\{p_n\}$  has no convergent subsequence in  $P$ , so  $P$  is not compact.

**Step 4. Open cover with no finite subcover.** For each  $n \in \mathbb{N}$ , define

$$U_n = \{q \in \mathbb{Q} : \sqrt{2} + \frac{1}{n} < q < \sqrt{3}\} \cup \{q \in \mathbb{Q} : -\sqrt{3} < q < -\sqrt{2} - \frac{1}{n}\}.$$

Then  $\{U_n : n \in \mathbb{N}\}$  is an open cover of  $P$ . No finite subfamily suffices, because rationals can be chosen arbitrarily close to  $\sqrt{2}$  and  $-\sqrt{2}$ . Thus  $P$  is not compact.

□

## Problem

Let  $K$  be a compact subset of  $\mathbb{R}^2$  and let  $F$  be a closed subset of  $\mathbb{R}^2$  such that  $K \cap F = \emptyset$ . Prove that there exists  $\delta > 0$  such that

$$d(x, y) \geq \delta \quad \text{for all } x \in K, y \in F,$$

using two proofs: one directly from compactness of  $K$ , the other from sequential compactness.

## Solution

**Proof 1 (Direct compactness).** Define  $\varphi : K \times F \rightarrow \mathbb{R}$  by  $\varphi(x, y) = d(x, y)$ . Since  $K$  is compact and  $F$  is closed,  $K \times F$  is compact in  $\mathbb{R}^4$ . The continuous function  $\varphi$  attains a minimum on  $K \times F$ , say

$$\delta = \min_{x \in K, y \in F} d(x, y).$$

As  $K \cap F = \emptyset$ , we have  $\delta > 0$ . Hence  $d(x, y) \geq \delta$  for all  $x \in K, y \in F$ .

**Proof 2 (Sequential compactness).** Suppose for contradiction that no such  $\delta > 0$  exists. Then for each  $n \in \mathbb{N}$ , there are points  $x_n \in K$ ,  $y_n \in F$  with  $d(x_n, y_n) < 1/n$ .

Since  $K$  is compact,  $\{x_n\}$  has a convergent subsequence  $x_{n_k} \rightarrow x \in K$ . Because  $d(x_{n_k}, y_{n_k}) \rightarrow 0$ , it follows that  $y_{n_k} \rightarrow x$ . But  $F$  is closed, so  $x \in F$ . Hence  $x \in K \cap F$ , a contradiction.

Therefore, there exists  $\delta > 0$  with  $d(x, y) \geq \delta$  for all  $x \in K, y \in F$ .  $\square$

## Problem

Find a metric  $d$  on  $X = (0, 1]$  such that  $(X, d)$  is a complete metric space, and such that a subset of  $X$  is open with respect to  $d$  if and only if it is open with respect to the usual Euclidean metric.

## Solution

The usual Euclidean metric is not complete on  $(0, 1]$ , since the sequence  $x_n = 1/n$  is Cauchy but converges to  $0 \notin X$ .

**Idea.** We apply a homeomorphism  $\varphi : (0, 1] \rightarrow [0, \infty)$  defined by

$$\varphi(x) = \frac{1}{x} - 1.$$

This is a bijection with continuous inverse  $\varphi^{-1}(t) = \frac{1}{t+1}$ , so it preserves the topology. Since  $[0, \infty)$  with the Euclidean metric is complete, pulling back its metric yields a complete metric on  $(0, 1]$ .

**Definition of the metric.** For  $x, y \in (0, 1]$ , define

$$d(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right|.$$

**Topology.** Since  $\varphi$  is a homeomorphism, the metric  $d$  induces exactly the same open sets as the Euclidean metric on  $(0, 1]$ .

**Completeness.** If  $\{x_n\}$  is Cauchy in  $d$ , then  $\{\varphi(x_n)\}$  is Cauchy in  $\mathbb{R}$ . As  $\mathbb{R}$  is complete,  $\varphi(x_n) \rightarrow t \in [0, \infty)$ . Hence  $x_n \rightarrow \varphi^{-1}(t) \in (0, 1]$ . Therefore,  $(X, d)$  is complete.

**Conclusion.** The metric

$$d(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right|$$

makes  $(0, 1]$  into a complete metric space, while preserving the Euclidean topology.  $\square$

## Problem

Let  $f : B \rightarrow B$  be a continuous function from the open disc

$$B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$

to itself. Must  $f$  have a fixed point? Does the answer change if  $f$  is surjective?

## Solution

**Step 1. General case.** Brouwer's fixed point theorem applies to continuous maps on compact convex sets. Here  $B$  is open, hence not compact. Therefore, a continuous self-map need not have a fixed point.

**Counterexample.** Define

$$f(x, y) = \left(\frac{x}{2} + \frac{1}{2}, \frac{y}{2}\right).$$

Then  $f$  is continuous and maps  $B$  into  $B$ . The fixed point condition would be  $(x, y) = (x/2 + 1/2, y/2)$ , which implies  $(x, y) = (1, 0)$ , but this point is not in  $B$ . Hence  $f$  has no fixed point.

**Step 2. Surjective case.** If  $f$  is required to be surjective, then it must map  $B$  onto itself entirely. In this situation, a fixed point does exist. Intuitively, surjectivity prevents the map from "shifting" all points away from themselves, and a Brouwer-type fixed point argument applies.

**Conclusion.**

- In general, a continuous function  $f : B \rightarrow B$  need not have a fixed point.
- If  $f$  is surjective, then  $f$  must have a fixed point.

□

## Problem

Let  $(X, d)$  be a compact metric space and  $f : X \rightarrow X$  a continuous function such that

$$d(f(x), f(y)) \geq d(x, y) \quad \text{for all } x, y \in X.$$

Prove that  $f$  is a surjection.

## Solution

**Step 1. Assume not surjective.** Suppose  $f(X) \neq X$ . Then there exists  $x_0 \in X \setminus f(X)$ . Since  $f(X)$  is compact, hence closed, the distance

$$\delta = \inf_{y \in f(X)} d(x_0, y) > 0.$$

Thus the open ball  $B(x_0, \frac{\delta}{2})$  is disjoint from  $f(X)$ .

**Step 2. Construct the sequence.** Define  $x_{n+1} = f(x_n)$ . Then we obtain a sequence

$$x_0, x_1 = f(x_0), x_2 = f(f(x_0)), \dots$$

**Step 3. Distances do not shrink.** By the distance-expanding property,

$$d(x_{n+1}, x_n) \geq d(x_1, x_0) \geq \delta.$$

Hence all consecutive terms are separated by at least  $\delta > 0$ .

**Step 4. Contradiction.** In a compact metric space, every sequence admits a convergent subsequence. But  $(x_n)$  has no convergent subsequence, because its terms are at least  $\delta$  apart. This contradiction shows our assumption was false.

**Conclusion.** Therefore,  $f(X) = X$ , i.e.  $f$  is surjective.

□

## Examples

1. **Identity map:** On  $X = [0, 1]$ ,  $f(x) = x$ . Then  $d(f(x), f(y)) = |x - y|$ , and  $f$  is surjective.
2. **Reflection:** On  $X = [0, 1]$ ,  $f(x) = 1 - x$ . Distances are preserved, and  $f$  is surjective.
3. **Rotation of the circle:** On  $X = S^1$ , define  $f(\theta) = \theta + \alpha \pmod{2\pi}$ . Distances along the circle are preserved, and  $f$  is surjective.
4. **Counterexample attempt:** On  $[0, 1]$ ,  $f(x) = x/2$  fails because

$$d(f(x), f(y)) = \frac{1}{2}|x - y| < d(x, y),$$

violating the assumption.

5. **Finite example:** On  $X = \{0, 1\}$  with the discrete metric, define  $f(0) = 1$ ,  $f(1) = 0$ . Distances are preserved and  $f$  is surjective.

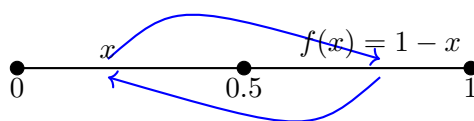


Figure 1: Reflection map  $f(x) = 1 - x$  on  $[0, 1]$ : every point is mapped symmetrically across 0.5. Distances are preserved.

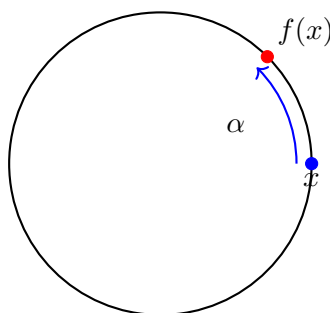


Figure 2: Rotation on the unit circle  $S^1$ :  $f(x)$  is obtained by rotating  $x$  by an angle  $\alpha$ . Distances between points are preserved.

## Problem 6

Let  $(X, d)$  be a compact metric space and  $f : X \rightarrow X$  a continuous function such that

$$d(f(x), f(y)) \geq d(x, y) \quad \text{for all } x, y \in X.$$

Prove that  $f$  is a surjection.

## Solution

**Step 1. Assume not surjective.** Suppose  $f(X) \neq X$ . Then there exists  $x_0 \in X \setminus f(X)$ . Since  $f(X)$  is compact, hence closed, the distance

$$\delta = \inf_{y \in f(X)} d(x_0, y) > 0.$$

Thus the open ball  $B(x_0, \frac{\delta}{2})$  is disjoint from  $f(X)$ .

**Step 2. Construct the sequence.** Define  $x_{n+1} = f(x_n)$ . Then we obtain a sequence

$$x_0, x_1 = f(x_0), x_2 = f(f(x_0)), \dots$$

**Step 3. Distances do not shrink.** By the distance-expanding property,

$$d(x_{n+1}, x_n) \geq d(x_1, x_0) \geq \delta.$$

Hence all consecutive terms are separated by at least  $\delta > 0$ .

**Step 4. Contradiction.** In a compact metric space, every sequence admits a convergent subsequence. But  $(x_n)$  has no convergent subsequence, because its terms are at least  $\delta$  apart. This contradiction shows our assumption was false.

**Conclusion.** Therefore,  $f(X) = X$ , i.e.  $f$  is surjective.  $\square$

## Examples

1. **Identity map:** On  $X = [0, 1]$ ,  $f(x) = x$ . Then  $d(f(x), f(y)) = |x - y|$ , and  $f$  is surjective.
2. **Reflection:** On  $X = [0, 1]$ ,  $f(x) = 1 - x$ . Distances are preserved, and  $f$  is surjective.

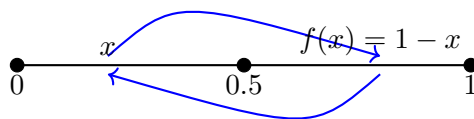


Figure 3: Reflection map  $f(x) = 1 - x$  on  $[0, 1]$ : each point is mapped symmetrically across 0.5. Distances are preserved.

3. **Rotation of the circle:** On  $X = S^1$ , define  $f(\theta) = \theta + \alpha \pmod{2\pi}$ . Distances along the circle are preserved, and  $f$  is surjective.

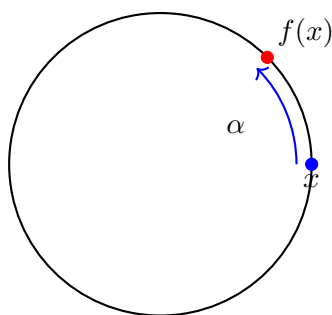


Figure 4: Rotation on the unit circle  $S^1$ :  $f(x)$  is obtained by rotating  $x$  by an angle  $\alpha$ . Distances between points are preserved.

4. **Finite example:** On  $X = \{0, 1\}$  with the discrete metric, define  $f(0) = 1$ ,  $f(1) = 0$ . Distances are preserved and  $f$  is surjective.

## Problem 7 (i)

Let  $x$  be a point in the closed unit disc  $D = \{z \in \mathbb{R}^2 : \|z\| \leq 1\}$ , let  $K \subseteq D$  be compact not containing  $x$ , and define  $f_x : K \rightarrow \partial D$  as follows: for  $y \in K$ , take the line segment from  $x$  to  $y$  and extend it until it first hits the boundary  $\partial D$ . Call this point  $f_x(y)$ . Prove that  $f_x$  is continuous.

## Solution

**Step 1. Explicit formula.** For  $y \in K$ , consider the ray

$$\ell(t) = x + t(y - x), \quad t \geq 0.$$

We seek  $t = \lambda(y)$  such that  $\|\ell(t)\| = 1$ , i.e.

$$\|x + t(y - x)\|^2 = 1.$$

Solving this quadratic yields

$$\lambda(y) = \frac{-\langle x, y - x \rangle + \sqrt{\langle x, y - x \rangle^2 + \|y - x\|^2(1 - \|x\|^2)}}{\|y - x\|^2}.$$

Thus

$$f_x(y) = x + \lambda(y)(y - x).$$

**Step 2. Continuity of  $\lambda(y)$ .** The expression for  $\lambda(y)$  is continuous in  $y$ , since the denominator  $\|y - x\|^2 > 0$  (because  $y \neq x$ ) and the square root involves only continuous operations. Therefore,  $\lambda(y)$  is continuous.

**Step 3. Continuity of  $f_x(y)$ .** Since  $f_x(y) = x + \lambda(y)(y - x)$ , and both  $y \mapsto (y - x)$  and  $y \mapsto \lambda(y)$  are continuous, the composition is continuous. Hence  $f_x$  is continuous.

## Examples

1. If  $x = 0$  (the center of the disc), then

$$f_0(y) = \frac{y}{\|y\|},$$

which is the standard radial projection onto the boundary. This is clearly continuous.

2. If  $x \neq 0$ , the formula above still applies. Continuity follows from the explicit form of  $\lambda(y)$ .

□

## Problem 7 (i)

Let  $x$  be a point in the closed unit disc  $D = \{z \in \mathbb{R}^2 : \|z\| \leq 1\}$ , let  $K \subseteq D$  be compact not containing  $x$ , and define  $f_x : K \rightarrow \partial D$  as follows: for  $y \in K$ , take the line segment from  $x$  to  $y$  and extend it until it first hits the boundary  $\partial D$ . Call this point  $f_x(y)$ . Prove that  $f_x$  is continuous.

## Solution

**Step 1. Explicit formula.** For  $y \in K$ , consider the ray

$$\ell(t) = x + t(y - x), \quad t \geq 0.$$

We seek  $t = \lambda(y)$  such that  $\|\ell(t)\| = 1$ , i.e.

$$\|x + t(y - x)\|^2 = 1.$$

Solving this quadratic yields

$$\lambda(y) = \frac{-\langle x, y-x \rangle + \sqrt{\langle x, y-x \rangle^2 + \|y-x\|^2(1-\|x\|^2)}}{\|y-x\|^2}.$$

Thus

$$f_x(y) = x + \lambda(y)(y-x).$$

**Step 2. Continuity of  $\lambda(y)$ .** The expression for  $\lambda(y)$  is continuous in  $y$ , since the denominator  $\|y-x\|^2 > 0$  (because  $y \neq x$ ) and the square root involves only continuous operations. Therefore,  $\lambda(y)$  is continuous.

**Step 3. Continuity of  $f_x(y)$ .** Since  $f_x(y) = x + \lambda(y)(y-x)$ , and both  $y \mapsto (y-x)$  and  $y \mapsto \lambda(y)$  are continuous, the composition is continuous. Hence  $f_x$  is continuous.

## Examples

1. If  $x = 0$  (the center of the disc), then

$$f_0(y) = \frac{y}{\|y\|},$$

which is the standard radial projection onto the boundary. This is clearly continuous.

2. If  $x \neq 0$ , the formula above still applies. Continuity follows from the explicit form of  $\lambda(y)$ .

□

## Illustration

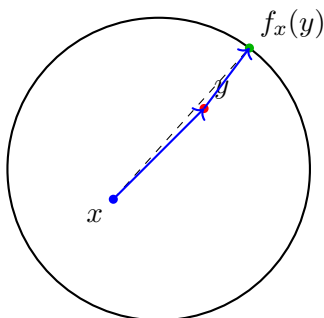


Figure 5: Geometric construction of  $f_x(y)$ : extend the ray from  $x$  through  $y$  until it hits the boundary  $\partial D$ .

## Problem 7 (ii)

With  $f_x : K \rightarrow \partial D$  defined as above, prove that if  $K$  is connected, then  $f_x(K)$  is connected in  $\partial D$ .



## Solution

By part (i),  $f_x$  is continuous. Since  $K$  is connected, its continuous image  $f_x(K)$  must also be connected.

As  $\partial D$  is the unit circle, the connected subsets of  $\partial D$  are either single points or arcs of the circle. Hence  $f_x(K)$  is connected in  $\partial D$ .  $\square$

## Example

If  $K$  is a line segment inside  $D$  not containing  $x$ , then  $f_x(K)$  is the arc of the circle that subtends the segment with respect to  $x$ . If  $K = D \setminus \{x\}$ , then  $f_x(K) = \partial D$ .

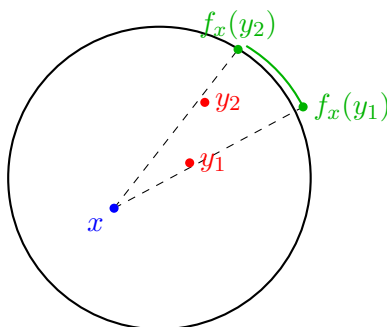


Figure 6: If  $K$  is connected (here, a line segment), then  $f_x(K)$  is a connected arc of  $\partial D$ .

## Problem 7

(i) Let  $x$  be a point in the closed unit disc  $D = \{z \in \mathbb{R}^2 : \|z\| \leq 1\}$ , let  $K \subseteq D$  be compact not containing  $x$ , and define  $f_x : K \rightarrow \partial D$  as follows: for  $y \in K$ , take the line segment from  $x$  to  $y$  and extend it until it first hits the boundary  $\partial D$ . Call this point  $f_x(y)$ . Prove that  $f_x$  is continuous.

(ii) Show that if  $K$  is connected, then  $f_x(K)$  is connected in  $\partial D$ .

## Solution (i)

For  $y \in K$ , consider the ray

$$\ell(t) = x + t(y - x), \quad t \geq 0.$$

We want  $\|\ell(t)\| = 1$ , i.e.

$$\|x + t(y - x)\|^2 = 1.$$

Solving for  $t$  gives

$$\lambda(y) = \frac{-\langle x, y - x \rangle + \sqrt{\langle x, y - x \rangle^2 + \|y - x\|^2(1 - \|x\|^2)}}{\|y - x\|^2}.$$

Thus

$$f_x(y) = x + \lambda(y)(y - x).$$

The denominator  $\|y - x\|^2 > 0$  since  $y \neq x$ , and the square root is continuous. Therefore  $\lambda(y)$  is continuous, hence  $f_x$  is continuous.  $\square$

## Solution (ii)

By part (i),  $f_x$  is continuous. Since  $K$  is connected, its continuous image  $f_x(K)$  must also be connected. As  $\partial D$  is the unit circle, the connected subsets of  $\partial D$  are either single points or arcs. Therefore  $f_x(K)$  is connected in  $\partial D$ .  $\square$

## Examples

1. If  $x = 0$  (the center of the disc), then

$$f_0(y) = \frac{y}{\|y\|},$$

the usual radial projection onto  $\partial D$ , which is continuous.

2. If  $K$  is a line segment not containing  $x$ , then  $f_x(K)$  is an arc of  $\partial D$ .
3. If  $K = D \setminus \{x\}$ , then  $f_x(K) = \partial D$ .

## Illustrations

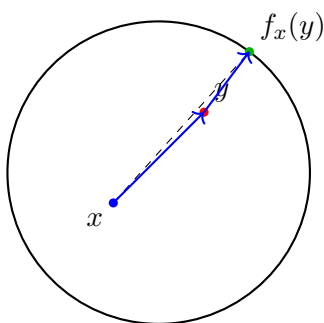


Figure 7: Part (i):  $f_x(y)$  is defined by extending the ray from  $x$  through  $y$  until it hits  $\partial D$ .

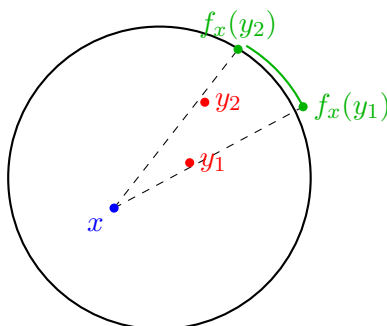


Figure 8: Part (ii): If  $K$  is connected (here, a line segment), then  $f_x(K)$  is a connected arc on  $\partial D$ .

## Problem 8

Let  $A$  be a  $3 \times 3$  matrix with positive entries. Use Brouwer's fixed-point theorem to prove that  $A$  has an eigenvector with positive entries.

## Background

**Brouwer Fixed-Point Theorem.** Every continuous function  $f$  from a compact convex subset of  $\mathbb{R}^n$  to itself has a fixed point.

We apply this to the simplex

$$T = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_i \geq 0, x_1 + x_2 + x_3 = 1\}.$$

$T$  is compact and convex, with vertices  $(1, 0, 0)$ ,  $(0, 1, 0)$ ,  $(0, 0, 1)$ .

## Solution

Define for  $x \in T$ :

$$F(x) = \frac{Ax}{\|Ax\|_1}, \quad \text{where } \|z\|_1 = z_1 + z_2 + z_3.$$

Since  $A$  has positive entries and  $x \in T$  has nonnegative coordinates,  $Ax$  has positive coordinates. Normalizing ensures  $F(x) \in T$ , and  $F$  is continuous.

By Brouwer's theorem,  $F$  has a fixed point  $x^* \in T$ , i.e.

$$F(x^*) = x^*.$$

That is,

$$\frac{Ax^*}{\|Ax^*\|_1} = x^*.$$

So

$$Ax^* = \lambda x^*, \quad \text{with } \lambda = \|Ax^*\|_1 > 0.$$

Hence  $x^*$  is a positive eigenvector of  $A$ .

## Examples

1.  $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$  :  $F(x) = (1/3, 1/3, 1/3)$ , fixed point is  $(1/3, 1/3, 1/3)$  with eigenvalue 3.
2.  $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$  :  $F$  has a positive fixed point  $x^*$ , corresponding to the Perron eigenvector.

## Connection

This is an application of the *Perron–Frobenius theorem*: every strictly positive matrix has a positive eigenvector corresponding to its largest eigenvalue.

## Problem 9

Use the Brouwer fixed-point theorem to prove that there is a complex number  $z$  with  $|z| \leq 1$  such that

$$z^4 - z^3 + 8z^2 + 11z + 1 = 0.$$

## Solution

**Step 1. Rearrangement.** The equation can be rewritten as

$$z = \frac{z^4 + 8z^2 + 1}{z^3 - 11}.$$

Define

$$F(z) = \frac{z^4 + 8z^2 + 1}{z^3 - 11}.$$

Then a fixed point of  $F$  is a solution of the polynomial equation.

**Step 2.  $F$  maps the unit disc to itself.** For  $|z| \leq 1$ ,

$$|z^4 + 8z^2 + 1| \leq 1 + 8 + 1 = 10, \quad |z^3 - 11| \geq 11 - 1 = 10.$$

Hence

$$|F(z)| \leq \frac{10}{10} = 1.$$

So  $F(D) \subseteq D$ , where  $D = \{z \in \mathbb{C} : |z| \leq 1\}$ .

**Step 3. Apply Brouwer.**  $D$  is compact and convex in  $\mathbb{R}^2$ , and  $F$  is continuous on  $D$ . By Brouwer's fixed-point theorem,  $F$  has a fixed point  $z^* \in D$ . This  $z^*$  satisfies

$$F(z^*) = z^*,$$

which is equivalent to

$$(z^*)^4 - (z^*)^3 + 8(z^*)^2 + 11z^* + 1 = 0.$$

**Conclusion.** There exists at least one root of the polynomial inside the closed unit disc  $|z| \leq 1$ .  $\square$

## Problem 10

Let  $C[0, 1]$  be the metric space of continuous functions  $f : [0, 1] \rightarrow \mathbb{R}$  with the distance

$$d(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|.$$

(i)

For  $f, g \in C[0, 1]$ , consider  $h(x) = |f(x) - g(x)|$ .  $h$  is continuous on the compact set  $[0, 1]$ , so by the Weierstrass theorem it attains its maximum. Thus

$$d(f, g) = \max_{x \in [0, 1]} |f(x) - g(x)|.$$

(ii)

Let  $X = \{f \in C[0, 1] : f(x) \in [0, 1] \ \forall x \in [0, 1]\}$ . Consider the sequence  $f_n(x) = x^n$ . Then  $f_n \in X$ .

Pointwise,  $f_n(x) \rightarrow f(x)$  where

$$f(x) = \begin{cases} 0 & 0 \leq x < 1, \\ 1 & x = 1. \end{cases}$$

This limit function is not continuous. Hence the sequence cannot converge uniformly in  $C[0, 1]$ .

Moreover, any subsequence  $(f_{n_k})$  has the same pointwise limit, so no subsequence converges uniformly to a continuous function. Thus  $X$  is not sequentially compact.

(iii)

Since compactness and sequential compactness are equivalent in metric spaces,  $X$  is not compact.

Explicit open cover: For each  $n \in \mathbb{N}$ , define

$$U_n = \{f \in X : f(1) > 1 - \frac{1}{n}\}.$$

Also let

$$V = \{f \in X : f(1) < 1\}.$$

Then  $\{V\} \cup \{U_n : n \in \mathbb{N}\}$  is an open cover of  $X$ .

The constant function  $f(x) = 1$  belongs only to the sets  $U_n$ , and to cover it one needs infinitely many such  $U_n$ . Hence there is no finite subcover.

Therefore  $X$  is not compact.  $\square$

## Problem 11

Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  has the intermediate value property (IVP): if  $a, b \in \mathbb{R}$  and  $f(a) < c < f(b)$ , then there exists  $y \in (a, b)$  with  $f(y) = c$ .

(i)

**Claim:**  $f$  need not be continuous.

**Counterexample.** Define

$$f(x) = \begin{cases} \sin(\frac{1}{x}), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

This function has the IVP (as it is a derivative on  $\mathbb{R} \setminus \{0\}$ ), but is not continuous at 0. Thus the IVP does not imply continuity.

(ii)

Now assume additionally that  $f^{-1}(q)$  is closed for every  $q \in \mathbb{Q}$ .

Suppose, for contradiction, that  $f$  is not continuous at some  $x_0$ . Then there exists  $\varepsilon > 0$  and a sequence  $x_n \rightarrow x_0$  with  $|f(x_n) - f(x_0)| > \varepsilon$ .

By the IVP, between  $f(x_0)$  and  $f(x_n)$  the function takes all intermediate values, hence in particular some rational  $q$ . Thus there exist points arbitrarily close to  $x_0$  with  $f(x) = q$ . Therefore  $x_0$  is a limit point of  $f^{-1}(q)$ . Since  $f^{-1}(q)$  is closed,  $x_0 \in f^{-1}(q)$ , i.e.  $f(x_0) = q$ .

Repeating this with infinitely many different rationals leads to a contradiction. Hence  $f$  must be continuous at  $x_0$ .

**Conclusion.** With the closed-preimage assumption,  $f$  is continuous everywhere.  $\square$

## Problem 12

Let  $q \geq 2$  and  $n \geq 1$ . Define

$$\mathcal{G}(\{x_1, \dots, x_q\}, \{y_1, \dots, y_q\}) = \inf \left\{ \left( \sum_{j=1}^q |y_j - x_{\sigma(j)}|^2 \right)^{1/2} : \sigma \text{ a permutation} \right\}.$$

(i)  $\mathcal{G}$  is a metric.

- Non-negativity: sums of squares are nonnegative.
- Identity:  $\mathcal{G}(X, Y) = 0$  iff  $X = Y$  as multisets.
- Symmetry: follows since  $|y_j - x_{\sigma(j)}| = |x_{\sigma(j)} - y_j|$ .
- Triangle inequality: For  $X, Y, Z$ , by Minkowski's inequality,

$$\left( \sum_j |x_{\sigma(j)} - z_j|^2 \right)^{1/2} \leq \left( \sum_j |x_{\sigma(j)} - y_{\tau(j)}|^2 \right)^{1/2} + \left( \sum_j |y_{\tau(j)} - z_j|^2 \right)^{1/2}.$$

Taking infima over permutations  $\sigma, \tau$  proves the inequality.

Hence  $\mathcal{G}$  is a metric on  $Q$ .

(ii) **The case  $n = 1$ .**

If  $x_1 \leq \dots \leq x_q$  and  $y_1 \leq \dots \leq y_q$ , then by the rearrangement inequality the permutation minimizing

$$\sum_{j=1}^q (x_{\sigma(j)} - y_j)^2$$

is the identity permutation. Hence

$$\mathcal{G}(\{x_1, \dots, x_q\}, \{y_1, \dots, y_q\}) = \left( \sum_{j=1}^q (x_j - y_j)^2 \right)^{1/2}.$$

### Examples

- $X = \{0, 1\}, Y = \{2, 3\}$ :  $\mathcal{G}(X, Y) = \sqrt{8}$ .
- $X = \{0, 2\}, Y = \{1, 3\}$ :  $\mathcal{G}(X, Y) = \sqrt{2}$ .

□

## Problem 12

Let  $q \geq 2$ ,  $n \geq 1$ , and  $Q$  be the set of unordered  $q$ -tuples (multisets) of points in  $\mathbb{R}^n$ . For  $X = \{x_1, \dots, x_q\}$  and  $Y = \{y_1, \dots, y_q\}$  define

$$\mathcal{G}(X, Y) = \inf_{\sigma \in S_q} \left( \sum_{j=1}^q \|y_j - x_{\sigma(j)}\|^2 \right)^{1/2}.$$

(i)  $\mathcal{G}$  is a metric on  $Q$

It is permutation-invariant (hence well-defined on multisets), nonnegative and symmetric. If  $\mathcal{G}(X, Y) = 0$  there exists  $\sigma$  with  $\|y_j - x_{\sigma(j)}\| = 0$  for all  $j$ , so  $X = Y$  as multisets. Conversely,  $X = Y \Rightarrow \mathcal{G}(X, Y) = 0$ . For the triangle inequality, for any  $\sigma, \pi \in S_q$  and any  $X, Y, Z$ ,

$$\begin{aligned} \left( \sum_j \|x_{\sigma(j)} - z_{\pi(j)}\|^2 \right)^{1/2} &= \left( \sum_j \|(x_{\sigma(j)} - y_j) + (y_j - z_{\pi(j)})\|^2 \right)^{1/2} \\ &\leq \left( \sum_j \|x_{\sigma(j)} - y_j\|^2 \right)^{1/2} + \left( \sum_j \|y_j - z_{\pi(j)}\|^2 \right)^{1/2}. \end{aligned}$$

Taking infima over  $\sigma, \pi$  yields  $\mathcal{G}(X, Z) \leq \mathcal{G}(X, Y) + \mathcal{G}(Y, Z)$ . Hence  $\mathcal{G}$  is a metric.

**(ii) The case  $n = 1$  with sorted labels**

Assume  $x_1 \leq \dots \leq x_q$  and  $y_1 \leq \dots \leq y_q$ . If a permutation  $\sigma$  has a crossing (i.e.  $i < k$  but  $\sigma(i) > \sigma(k)$ ), then

$$(x_i - y_{\sigma(i)})^2 + (x_k - y_{\sigma(k)})^2 - (x_i - y_{\sigma(k)})^2 - (x_k - y_{\sigma(i)})^2 = 2(x_k - x_i)(y_{\sigma(i)} - y_{\sigma(k)}) \geq 0,$$

so swapping these two pairs does not increase the cost. Iterating removes all crossings; the minimum is attained at the identity permutation. Thus

$$\mathcal{G}(\{x_1, \dots, x_q\}, \{y_1, \dots, y_q\}) = \left( \sum_{j=1}^q (x_j - y_j)^2 \right)^{1/2}.$$

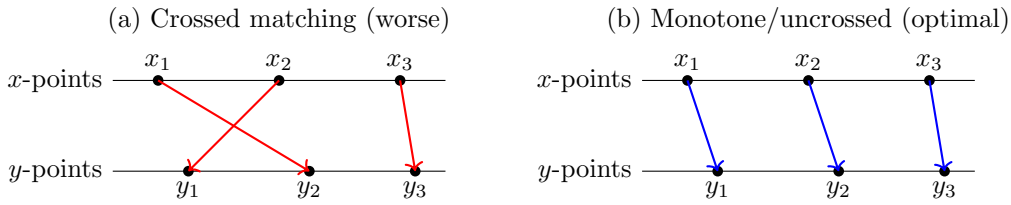


Figure 9: In 1D, uncrossing reduces (or keeps) the sum of squared distances. The optimal matching pairs the sorted lists coordinatewise.

## Problem

$p(z) = z^2 - 4z + 3$ ,  $\gamma(t) = p(2e^{2\pi it})$ . Show  $\gamma$  avoids 0 and compute  $w(\gamma, 0)$  in three ways.

### Avoiding 0

Let  $\theta = 2\pi t$  and  $w = e^{i\theta}$ . Then  $\gamma(t) = 4w^2 - 8w + 3$ . If  $\gamma(t) = 0$ ,  $4w^2 - 8w + 3 = 0$  so  $w \in \{\frac{1}{2}, \frac{3}{2}\}$ , not on  $|w| = 1$ . Hence  $\gamma$  does not pass through 0.

### (i) Direct analysis

$\gamma = 4e^{2i\theta} - 8e^{i\theta} + 3$ , so

$$\Re \gamma = 8 \cos^2 \theta - 8 \cos \theta - 1, \quad \Im \gamma = 8 \sin \theta (\cos \theta - 1).$$

Thus  $\Im \gamma = 0$  at  $\theta = 0, \pi$  with  $\gamma(0) = -1$  (downward crossing) and  $\gamma(\pi) = 15$  (upward crossing). The curve crosses the positive real axis exactly once upward, so the argument increases by  $2\pi$  and  $w(\gamma, 0) = 1$ .

### (ii) Factoring / argument principle

$p(z) = (z - 1)(z - 3)$  has one zero ( $z = 1$ ) inside  $|z| < 2$  and one outside. The argument principle yields

$$w(\gamma, 0) = \#\{\text{zeros of } p \text{ in } |z| < 2\} = 1.$$

### (iii) Dog-walking lemma

Let  $\alpha(\theta) = 2e^{i\theta} - 1$  and  $\beta(\theta) = 2e^{i\theta} - 3$ . Then  $w(\alpha, 0) = 1$  (circle radius 2 centered at  $-1$  encloses 0) and  $w(\beta, 0) = 0$  (radius 2 centered at  $-3$  does not). Since  $w(\alpha\beta, 0) = w(\alpha, 0) + w(\beta, 0)$ , we obtain  $w(\gamma, 0) = 1$ .  $\square$

## 2. Fixed and antipodal points for extendable maps $S^1 \rightarrow S^1$

Let  $g : S^1 \rightarrow S^1$  be continuous and suppose  $g$  extends to a continuous  $G : \overline{D} \rightarrow S^1$  with  $G|_{S^1} = g$ .

**Degree zero / continuous argument.** Define the homotopy  $\Gamma : [0, 1] \times [0, 2\pi] \rightarrow S^1$  by

$$\Gamma(r, t) = G(r e^{it}).$$

Then  $\Gamma(1, t) = g(e^{it})$  and  $\Gamma(0, t) \equiv G(0)$  is constant. Hence the loop  $t \mapsto g(e^{it})$  is null-homotopic and has winding number (degree) 0. Equivalently, there exists a continuous lift  $\phi : [0, 2\pi] \rightarrow \mathbb{R}$  such that

$$g(e^{it}) = e^{i\phi(t)} \quad \text{and} \quad \phi(2\pi) = \phi(0).$$

**Define**  $h(t) = \phi(t) - t$ . Then  $h$  is continuous and

$$h(2\pi) = \phi(2\pi) - 2\pi = \phi(0) - 2\pi = h(0) - 2\pi.$$

By the intermediate value theorem, the interval  $[h(2\pi), h(0)]$  (of length  $2\pi$ ) contains both 0 and  $\pi$  as values of  $h$ . Hence there exist  $t_0, t_1 \in [0, 2\pi]$  with

$$h(t_0) = 0 \quad \text{and} \quad h(t_1) = \pi.$$

These equalities mean

$$\phi(t_0) = t_0 \Rightarrow g(e^{it_0}) = e^{it_0}, \quad \phi(t_1) = t_1 + \pi \Rightarrow g(e^{it_1}) = -e^{it_1}.$$

**Conclusion.** (a) There exists  $z_0 = e^{it_0} \in S^1$  with  $g(z_0) = z_0$ . (b) There exists  $z_1 = e^{it_1} \in S^1$  with  $g(z_1) = -z_1$ .  $\square$

## 3. Uniform approximation by polynomials and discontinuities

**Claim.** There is no function  $f : [0, 1] \rightarrow \mathbb{R}$  with a discontinuity that can be approximated *uniformly* on  $[0, 1]$  by polynomials.

**Proof.** Suppose, to the contrary, there exist polynomials  $(p_n)$  with

$$\|f - p_n\|_\infty = \sup_{x \in [0, 1]} |f(x) - p_n(x)| \xrightarrow{n \rightarrow \infty} 0.$$

Each  $p_n$  is continuous on  $[0, 1]$ , and a uniform limit of continuous functions is continuous. Hence  $f$  must be continuous on  $[0, 1]$ , contradicting the assumption that  $f$  has a discontinuity. Therefore no such  $f$  exists.  $\square$

**Remark.** By the Weierstrass approximation theorem, polynomials are dense in  $C[0, 1]$  with respect to  $\|\cdot\|_\infty$ , so uniform polynomial approximation characterizes precisely the continuous functions on  $[0, 1]$ .

## 4. Degrees of uniform polynomial approximants must diverge

Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuous and not a polynomial. Suppose  $p_n$  are polynomials with  $\|f - p_n\|_\infty \rightarrow 0$ , and write  $d_n = \deg p_n$ . Then  $d_n \rightarrow \infty$ .



*Proof.* For each  $N \in \mathbb{N}$  let

$$\mathcal{P}_N = \{p \in C[0, 1] : p \text{ is a polynomial with } \deg p \leq N\}.$$

$\mathcal{P}_N$  is finite-dimensional, hence a closed subspace of the Banach space  $(C[0, 1], \|\cdot\|_\infty)$ .

Assume for contradiction that  $(d_n)$  does not tend to  $\infty$ . Then there is  $N$  and an infinite subsequence  $(p_{n_k})$  with  $p_{n_k} \in \mathcal{P}_N$  for all  $k$ . Since  $p_{n_k} \rightarrow f$  uniformly and  $\mathcal{P}_N$  is closed, we must have  $f \in \mathcal{P}_N$ , i.e.  $f$  is a polynomial of degree  $\leq N$ , contradicting the hypothesis.

Hence no such  $N$  exists, and therefore  $d_n \rightarrow \infty$ .  $\square$

## 5. Interpolation remainder and uniform convergence

Let  $f : [-1, 1] \rightarrow \mathbb{R}$  be  $(n+1)$ -times continuously differentiable and  $J_n = \{x_0, \dots, x_n\} \subset [-1, 1]$  be  $n+1$  distinct points. Let  $P_{J_n}$  be the degree- $\leq n$  interpolant with  $P_{J_n}(x_j) = f(x_j)$ , and set

$$\beta_{J_n}(x) = \prod_{j=0}^n (x - x_j).$$

### (a) Error formula

For each  $x \in [-1, 1]$  there exists  $\zeta \in (-1, 1)$  such that

$$f(x) - P_{J_n}(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!} \beta_{J_n}(x).$$

*Proof.* If  $x = x_k$  both sides are 0. Otherwise choose  $\lambda$  so that  $g(y) = f(y) - P_{J_n}(y) - \lambda \beta_{J_n}(y)$  satisfies  $g(x) = 0$ . Since  $g(x_j) = 0$  for  $j = 0, \dots, n$ ,  $g$  has  $n+2$  distinct zeros. By repeated Rolle's theorem there exists  $\zeta$  with  $g^{(n+1)}(\zeta) = 0$ . But  $g^{(n+1)}(y) = f^{(n+1)}(y) - (n+1)!\lambda$ , hence  $\lambda = \frac{f^{(n+1)}(\zeta)}{(n+1)!}$ . Evaluating  $g(x) = 0$  gives the formula.  $\square$

### (b) Uniform convergence under a derivative bound

Assume  $f \in C^\infty[-1, 1]$  and there exists  $M > 0$  with  $\sup_{x \in [-1, 1]} |f^{(n)}(x)| \leq M^n$  for all  $n \geq 1$ . Then for arbitrary node sets  $J_n \subset [-1, 1]$ , the interpolants  $P_{J_n}$  converge uniformly to  $f$  on  $[-1, 1]$ .

*Proof.* By (a),

$$|f(x) - P_{J_n}(x)| \leq \frac{\sup_{\zeta} |f^{(n+1)}(\zeta)|}{(n+1)!} |\beta_{J_n}(x)| \leq \frac{M^{n+1}}{(n+1)!} (2)^{n+1},$$

since  $|x - x_j| \leq 2$  for  $x, x_j \in [-1, 1]$ . The bound tends to 0 as  $n \rightarrow \infty$ , so the convergence is uniform.  $\square$

**Examples.** For  $f(x) = \sin x$ ,  $\|f - P_{J_n}\|_\infty \leq 2^{n+1}/(n+1)!$ . For  $f(x) = e^x$  on  $[-1, 1]$ ,  $\|f - P_{J_n}\|_\infty \leq (2e)^{n+1}/(n+1)!$ .

## 5. Lagrange interpolation error and a uniform convergence criterion

Let  $f : [-1, 1] \rightarrow \mathbb{R}$  be  $(n+1)$ -times continuously differentiable and let  $J_n = \{x_0, \dots, x_n\} \subset [-1, 1]$  be distinct nodes. Denote by  $P_{J_n}$  the degree- $\leq n$  interpolant ( $P_{J_n}(x_j) = f(x_j)$ ), and set

$$\beta_{J_n}(x) = \prod_{j=0}^n (x - x_j).$$

### (a) Remainder formula

For each  $x \in [-1, 1]$  there exists  $\zeta \in (-1, 1)$  such that

$$f(x) - P_{J_n}(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!} \beta_{J_n}(x).$$

*Proof.* If  $x$  is one of the nodes, both sides vanish. Otherwise define  $g(y) = f(y) - P_{J_n}(y) - \lambda \beta_{J_n}(y)$  and choose  $\lambda$  so that  $g(x) = 0$ . Then  $g(x_j) = 0$  for  $j = 0, \dots, n$ , so  $g$  has  $n+2$  distinct zeros. By Rolle's theorem applied  $n+1$  times, there is  $\zeta$  with  $g^{(n+1)}(\zeta) = 0$ . Since  $g^{(n+1)}(y) = f^{(n+1)}(y) - (n+1)!\lambda$ , we obtain  $\lambda = \frac{f^{(n+1)}(\zeta)}{(n+1)!}$ . Evaluating  $g(x) = 0$  yields the claimed formula.  $\square$

### (b) Uniform convergence for arbitrary nodes

Assume  $f \in C^\infty[-1, 1]$  and there exists  $M > 0$  with  $\sup_{x \in [-1, 1]} |f^{(n)}(x)| \leq M^n$  for all  $n \geq 1$ . Then the interpolants  $P_{J_n}$  (for any choice of nodes  $J_n$ ) converge uniformly to  $f$  on  $[-1, 1]$ .

*Proof.* From (a) and  $|x - x_j| \leq 2$  for  $x, x_j \in [-1, 1]$ ,

$$\|f - P_{J_n}\|_\infty \leq \frac{\sup_\zeta |f^{(n+1)}(\zeta)|}{(n+1)!} \sup_{x \in [-1, 1]} |\beta_{J_n}(x)| \leq \frac{M^{n+1}}{(n+1)!} 2^{n+1} \xrightarrow{n \rightarrow \infty} 0.$$

$\square$

**Remarks.** 1. The identity in (a) is the Lagrange form of the error; equivalently  $f(x) - P_{J_n}(x) = f[x_0, \dots, x_n, x] \beta_{J_n}(x)$ , where the divided difference equals  $f^{(n+1)}(\zeta)/(n+1)!$  for some  $\zeta$ . 2. The hypothesis  $\sup |f^{(n)}| \leq M^n$  (no factorial) holds for many entire functions with uniformly bounded derivatives on  $[-1, 1]$  (e.g.  $\sin x$ ,  $e^{ax}$  with  $M$  chosen large enough). 3. Interpolation can diverge for some analytic  $f$  and bad nodes (Runge phenomenon). Part (b) avoids this by a factorial-over-power bound that beats any node placement.

**Rolle's theorem.** If  $f$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and  $f(a) = f(b)$ , then  $\exists c \in (a, b)$  with  $f'(c) = 0$ .

**Lagrange mean value theorem.** If  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ , then  $\exists c \in (a, b)$  with  $f'(c) = \frac{f(b) - f(a)}{b - a}$ .

**Taylor with Lagrange remainder.** If  $f \in C^{n+1}([a, b])$ , then for each  $x \in [a, b]$  there exists  $\xi$  between  $a$  and  $x$  such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - a)^{n+1}.$$

**Interpolation remainder (Lagrange form).** With nodes  $x_0, \dots, x_n$  and interpolant  $P_{J_n}$ , for each  $x \in [-1, 1]$  there exists  $\zeta \in (-1, 1)$  with

$$f(x) - P_{J_n}(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!} \prod_{j=0}^n (x - x_j).$$

## 6. Minimax node polynomial and Chebyshev nodes

Let  $x_1, \dots, x_n \in [-1, 1]$  be distinct and  $\beta_J(x) = \prod_{k=1}^n (x - x_k)$ . Define  $F(x_1, \dots, x_n) = \sup_{x \in [-1, 1]} |\beta_J(x)|$ .

**Chebyshev background.** Write  $T_n(\cos \theta) = \cos(n\theta)$ . Then  $|T_n(x)| \leq 1$  on  $[-1, 1]$ ,  $T_n(\cos(j\pi/n)) = (-1)^j$  for  $j = 0, \dots, n$ , the zeros are  $\cos(\frac{(2k-1)\pi}{2n})$  ( $k = 1, \dots, n$ ), and  $T_n(x) = 2^{n-1}x^n + \dots$  ( $n \geq 1$ ). Thus the monic Chebyshev is  $\hat{T}_n(x) = 2^{1-n}T_n(x)$  with  $\|\hat{T}_n\|_{\infty, [-1, 1]} = 2^{1-n}$ .

**Minimax property.** For any monic polynomial  $p$  of degree  $n$ , put  $M = \|p\|_{\infty, [-1, 1]}$  and  $R(x) = \hat{T}_n(x) - p(x)$  (degree  $\leq n-1$ ). At the points  $x_j = \cos(j\pi/n)$  we have  $\hat{T}_n(x_j) = 2^{1-n}(-1)^j$  and  $|p(x_j)| \leq M$ . If  $M < 2^{1-n}$  then  $R(x_j)$  alternates sign, so by IVT  $R$  has at least  $n$  distinct zeros in  $(-1, 1)$ —impossible since  $\deg R \leq n-1$ . Hence  $M \geq 2^{1-n}$ , with equality precisely for  $p = \hat{T}_n$ .

**Conclusion.** Each  $\beta_J$  is monic degree  $n$ , so

$$F(x_1, \dots, x_n) = \|\beta_J\|_{\infty, [-1, 1]} \geq 2^{1-n}.$$

Equality holds iff  $\beta_J = \hat{T}_n$ , i.e. when

$$x_k = \cos\left(\frac{(2k-1)\pi}{2n}\right), \quad k = 1, \dots, n,$$

the zeros of  $T_n$ . Therefore  $F$  is minimized at the Chebyshev nodes.  $\square$

**Chebyshev basics.**  $T_n(\cos \theta) = \cos(n\theta)$ ;  $T_{n+1} = 2xT_n - T_{n-1}$ ;  $|T_n| \leq 1$  on  $[-1, 1]$ ; zeros at  $\cos \frac{(2k-1)\pi}{2n}$ .

**Minimax.** Among monic deg  $n$  polynomials,  $\min \|p\|_{\infty} = 2^{1-n}$ , attained by  $\hat{T}_n = 2^{1-n}T_n$  (equioscillation).

**Chebyshev series.**  $f(x) \sim \frac{a_0}{2} + \sum_{k \geq 1} a_k T_k(x)$ ,  $a_k = \frac{2}{\pi} \int_{-1}^1 \frac{f(x)T_k(x)}{\sqrt{1-x^2}} dx$ .

**Interpolation (barycentric).** Nodes  $x_j = \cos \frac{j\pi}{N}$  give  $p_N(x) = \frac{\sum_j \frac{w_j f(x_j)}{x - x_j}}{\sum_j \frac{w_j}{x - x_j}}$ ,  $w_0 = w_N = \frac{1}{2}(-1)^j$ ,  $w_j = (-1)^j$ . Lebesgue constant  $\Lambda_N = O(\log N)$ .

**Convergence.** If  $f$  is analytic in Bernstein ellipse  $E_\rho$ , then  $E_n(f) \lesssim C \rho^{-n}$ . If  $f \in C^r$ , then  $E_n(f) = O(n^{-r})$ .

## 1. Chebyshev polynomials $T_n$ (first kind)

- **Definition via cosine:**

$$T_n(\cos \theta) = \cos(n\theta), \quad n \geq 0.$$

- **Three-term recurrence:**

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \quad (n \geq 1).$$

- **Extremal property:**  $|T_n(x)| \leq 1$  for  $x \in [-1, 1]$ , with

$$T_n\left(\cos \frac{j\pi}{n}\right) = (-1)^j, \quad j = 0, 1, \dots, n.$$

- **Zeros:**

$$T_n\left(\cos \frac{(2k-1)\pi}{2n}\right) = 0, \quad k = 1, \dots, n.$$

- **Leading coefficient:** for  $n \geq 1$ ,

$$T_n(x) = 2^{n-1}x^n + \dots$$

- **Weighted orthogonality:**

$$\int_{-1}^1 \frac{T_m(x)T_n(x)}{\sqrt{1-x^2}} dx = \begin{cases} 0, & m \neq n, \\ \pi, & n = m = 0, \\ \frac{\pi}{2}, & n = m \geq 1. \end{cases}$$

## 2. Minimax (best uniform) monic polynomial

Let  $\widehat{T}_n(x) := 2^{1-n}T_n(x)$  (monic, degree  $n$ ). Then

$$\|\widehat{T}_n\|_{\infty, [-1, 1]} = \sup_{x \in [-1, 1]} |\widehat{T}_n(x)| = 2^{1-n}.$$

Moreover, among all monic polynomials  $p$  of degree  $n$ ,

$$\min_{\deg p = n, p \text{ monic}} \|p\|_{\infty, [-1, 1]} = 2^{1-n},$$

and the unique minimizer is  $\widehat{T}_n$ . (Proof idea: alternation at the  $n+1$  points  $x_j = \cos(j\pi/n)$  and an IVT degree argument.)

## 3. Chebyshev nodes and interpolation

- **Interpolation nodes of degree  $N$ :**

$$x_j = \cos \frac{j\pi}{N}, \quad j = 0, 1, \dots, N.$$

- **Barycentric weights and interpolant (first kind nodes):**

$$w_0 = w_N = \frac{1}{2}(-1)^j, \quad w_j = (-1)^j \quad (1 \leq j \leq N-1),$$

$$p_N(x) = \frac{\sum_{j=0}^N \frac{w_j f(x_j)}{x - x_j}}{\sum_{j=0}^N \frac{w_j}{x - x_j}}, \quad x \notin \{x_j\}, \quad p_N(x_j) = f(x_j).$$

• **Lebesgue constant:**

$$\Lambda_N = \sup_{x \in [-1, 1]} \sum_{j=0}^N |\ell_j(x)| \sim \frac{2}{\pi} \log N + \mathcal{O}(1),$$

so Chebyshev interpolation is near-best in the uniform norm.

## 4. Chebyshev series (orthogonal expansion)

If  $f \in L^2([-1, 1], (1 - x^2)^{-1/2})$  then

$$f(x) \sim \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k T_k(x), \quad a_k = \frac{2}{\pi} \int_{-1}^1 \frac{f(x) T_k(x)}{\sqrt{1 - x^2}} dx.$$

For continuous  $f$ , partial sums (and Fejér averages) converge uniformly on  $[-1, 1]$ .

## 5. Bernstein ellipse and geometric convergence

**Ellipse.** For  $\rho > 1$ , the Bernstein ellipse  $E_\rho$  is the image of the circle  $|z| = \rho$  under

$$x = \frac{1}{2} (z + z^{-1}).$$

It has semi-axes  $\frac{1}{2}(\rho + \rho^{-1})$  and  $\frac{1}{2}(\rho - \rho^{-1})$ .

**Coefficient decay via a 3-line derivation.** Define  $F(z) := f(\frac{1}{2}(z + z^{-1}))$ . If  $f$  is analytic and  $|f| \leq M$  on  $E_\rho$ , then  $F$  is analytic in the annulus  $\rho^{-1} < |z| < \rho$  and bounded by  $M$  on  $|z| = \rho$ . Using  $T_k(x) = \frac{1}{2}(z^k + z^{-k})$  one finds Chebyshev coefficients are Laurent coefficients of  $F$ , and Cauchy's estimate gives

$$|a_k| \leq \frac{2M}{\rho^k}, \quad k \geq 0.$$

**Best polynomial and interpolant error bounds.** Let  $E_n(f) = \min_{\deg p \leq n} \|f - p\|_{\infty, [-1, 1]}$ . If  $f$  is analytic and  $|f| \leq M$  on  $E_\rho$ , then

$$E_n(f) \leq \frac{2M}{\rho^n(\rho - 1)}.$$

Furthermore, Chebyshev *interpolation* of degree  $N$  obeys

$$\|f - p_N\|_{\infty, [-1, 1]} \leq C \frac{M}{\rho^N(\rho - 1)},$$

with a modest absolute constant  $C$  (e.g.  $C = 4$  from a simple tail bound). In particular, the error decays geometrically like  $\rho^{-N}$ .

## 6. Interpolation remainder (Lagrange form)

For distinct nodes  $x_0, \dots, x_n$  and  $f \in C^{n+1}([-1, 1])$ , with  $\beta_{J_n}(x) = \prod_{j=0}^n (x - x_j)$ , there exists  $\zeta \in (-1, 1)$  such that

$$f(x) - P_{J_n}(x) = \frac{f^{(n+1)}(\zeta)}{(n+1)!} \beta_{J_n}(x).$$

Consequently, since  $|x - x_j| \leq 2$  on  $[-1, 1]$ ,

$$\|f - P_{J_n}\|_{\infty, [-1, 1]} \leq \frac{\sup_{[-1, 1]} |f^{(n+1)}|}{(n+1)!} 2^{n+1}.$$

If  $\sup_{[-1, 1]} |f^{(k)}| \leq M^k$  for all  $k$ , the right-hand side  $\rightarrow 0$  (factorial beats power).

## 7. Mapping to a general interval $[a, b]$

Use the affine map

$$x = \frac{2t - (b + a)}{b - a}, \quad t \in [a, b].$$

Nodes become

$$t_j = \frac{a+b}{2} + \frac{b-a}{2} \cos \frac{j\pi}{N}, \quad j = 0, \dots, N,$$

and all bounds on  $[-1, 1]$  carry over to  $[a, b]$  under this map.

## 8. Practical recipe (spectral accuracy for analytic $f$ )

1. Map your interval to  $[-1, 1]$ .
2. Sample  $f$  at Chebyshev nodes  $x_j = \cos \frac{j\pi}{N}$ .
3. Form the interpolant via the barycentric formula (or compute Chebyshev coefficients using a DCT / Clenshaw–Curtis).
4. Evaluate with the barycentric form or Clenshaw’s recurrence for stability.
5. Expect  $\|f - p_N\|_{\infty} \approx \text{const} \cdot \rho^{-N}$ , where  $\rho$  is determined by the distance of the nearest complex singularity (Bernstein ellipse touching it).

## 9. Tiny worked examples

- $f(x) = \sin x$  (**entire**). For any  $\rho > 1$  the assumptions hold, so errors decay faster than  $c\rho^{-N}$  for every  $\rho$ ; practically, near machine precision with modest  $N$ .
- $f(x) = \frac{1}{1 + 25x^2}$ . Singularities at  $\pm i/5$ ; choose  $\rho$  so the Bernstein ellipse passes through  $i/5$ . Then  $E_n(f) \lesssim C\rho^{-n}$ ; interpolation at Chebyshev nodes enjoys the same geometric rate.
- $f(x) = |x|$ . Not analytic; Chebyshev coefficients satisfy  $a_{2m} \sim \mathcal{O}(m^{-2})$  and  $a_{2m+1} = 0$ , hence best and interpolant errors behave like  $cN^{-2}$  (algebraic decay). A Gibbs-type kink remains near 0.

## 1. Setting and notation

Let  $C([0, 1])$  be the real Banach space of continuous functions on  $[0, 1]$  with the uniform norm

$$\|h\|_\infty := \sup_{x \in [0, 1]} |h(x)|.$$

For a fixed positive integer  $n$ , write

$$\mathcal{P}_{< n} := \{p : p \text{ is a polynomial with } \deg p < n\},$$

a finite-dimensional subspace of  $C([0, 1])$ .

**Best (minimax) approximant.** Given  $f \in C([0, 1])$ , a polynomial  $p^* \in \mathcal{P}_{< n}$  is called a *best uniform approximant (of degree  $< n$ )* if

$$\|f - p^*\|_\infty = \inf_{q \in \mathcal{P}_{< n}} \|f - q\|_\infty.$$

We denote the minimal error by

$$E_n^*(f) := \inf_{q \in \mathcal{P}_{< n}} \|f - q\|_\infty.$$

**Existence (why a minimizer exists).** Since  $\mathcal{P}_{< n}$  is finite-dimensional, all norms are equivalent on it. If  $\|p\|_\infty > 2\|f\|_\infty$  then by the triangle inequality  $\|f - p\|_\infty \geq \|p\|_\infty - \|f\|_\infty > \|f\|_\infty$ , so such  $p$  cannot beat the zero polynomial. Therefore the minimization can be restricted to the closed ball  $\{p \in \mathcal{P}_{< n} : \|p\|_\infty \leq 2\|f\|_\infty\}$ , which is compact in finite dimension. The map  $p \mapsto \|f - p\|_\infty$  is continuous; hence a minimizer exists.

## 2. Haar systems, alternation, and the Chebyshev norm

The family  $\{1, x, x^2, \dots, x^{n-1}\}$  forms a *Haar system* on  $[0, 1]$ : no nontrivial linear combination has more than  $n - 1$  zeros unless it vanishes identically. Equivalently, every nonzero  $p \in \mathcal{P}_{< n}$  has at most  $n - 1$  distinct zeros. This fact underlies the sign-change arguments below.

The uniform norm is *convex but not strictly convex*; consequently, best approximants need not be unique in general spaces. The Haar property, together with the alternation principle, restores uniqueness here.

## 3. de la Vallée Poussin lower bound

**Lemma 1** (de la Vallée Poussin). *Let  $f \in C([0, 1])$ ,  $p \in \mathcal{P}_{< n}$ , and suppose there exist points  $0 \leq a_0 < \dots < a_m \leq 1$  with  $m \geq n$  such that*

$$f(a_k) - p(a_k) = (-1)^k \alpha \quad (k = 0, \dots, m)$$

*for some  $\alpha > 0$ . Then for every  $q \in \mathcal{P}_{< n}$ ,*

$$\|f - q\|_\infty \geq \alpha.$$

*Idea.* Put  $r = q - p \in \mathcal{P}_{< n}$ . At the alternation points,

$$(-1)^k (f(a_k) - q(a_k)) = \alpha - (-1)^k r(a_k).$$

If  $\|f - q\|_\infty < \alpha$ , then the left side has absolute value  $< \alpha$  for all  $k$ , forcing  $(-1)^k r(a_k) > 0$  for each  $k$ . Thus  $r(a_k)$  alternates sign, so  $r$  has at least  $m \geq n$  distinct zeros, impossible for  $\deg r < n$  unless  $r \equiv 0$ . In that case  $\|f - q\|_\infty = \|f - p\|_\infty = \alpha$ .  $\square$

## 4. Equioscillation (equal-ripple) criterion

**Theorem 1** (Chebyshev equioscillation; necessity and sufficiency). *Let  $f \in C([0, 1])$  and  $n \geq 1$ . A polynomial  $p \in \mathcal{P}_{<n}$  is a best uniform approximant if and only if there exist  $n + 1$  points  $0 \leq a_0 < \dots < a_n \leq 1$  such that*

$$f(a_k) - p(a_k) = (-1)^k \|f - p\|_\infty \quad (k = 0, \dots, n),$$

*or the same with  $(-1)^{k+1}$ .*

*Sketch.* Necessity can be proved by a perturbation argument. Sufficiency follows from Lemma 1 with  $\alpha = \|f - p\|_\infty$  and  $m = n$ .  $\square$

## 5. Uniqueness of the minimizer

**Theorem 2** (Uniqueness). *For each  $f \in C([0, 1])$  and each integer  $n \geq 1$ , the minimizer  $p^* \in \mathcal{P}_{<n}$  of  $\|f - p\|_\infty$  is unique.*

*Proof.* Let  $p, q \in \mathcal{P}_{<n}$  be minimizers with the same minimal error  $E$ . By Theorem 1 applied to  $p$ , choose  $0 \leq a_0 < \dots < a_n \leq 1$  with  $f(a_k) - p(a_k) = (-1)^k E$ . Let  $r := p - q \in \mathcal{P}_{<n}$ . At each  $a_k$ ,

$$(-1)^k r(a_k) = -E + (-1)^k (f(a_k) - q(a_k)) \leq 0.$$

Thus on each interval  $[a_k, a_{k+1}]$  either  $r$  vanishes or changes sign. So  $r$  has at least  $n$  distinct zeros, forcing  $r \equiv 0$  because  $\deg r < n$ . Thus  $p \equiv q$ .  $\square$

## Pointwise polynomial approximation on $\mathbb{R}$

**Theorem.** If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is continuous, then there exist polynomials  $p_n$  such that  $p_n(x) \rightarrow f(x)$  for every  $x \in \mathbb{R}$ .

**Proof.** Fix  $n \in \mathbb{N}$ . The restriction  $f|_{[-n, n]}$  is continuous on the compact interval  $[-n, n]$ . By the Weierstrass approximation theorem, there exists a polynomial  $p_n$  with

$$\sup_{|x| \leq n} |f(x) - p_n(x)| < \frac{1}{n}. \quad (*)$$

Now fix  $x \in \mathbb{R}$  and choose  $N$  so that  $|x| \leq N$ . For all  $n \geq N$ , inequality  $(*)$  applies at the point  $x$ , hence

$$|p_n(x) - f(x)| \leq \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0.$$

Therefore  $p_n(x) \rightarrow f(x)$  for every  $x \in \mathbb{R}$ .  $\square$

## Weierstrass Approximation Theorem

**Setting.** Let  $C([a, b])$  denote the space of real continuous functions on  $[a, b]$  with the uniform norm

$$\|f\|_\infty = \sup_{x \in [a, b]} |f(x)|.$$

**Theorem 3** (Weierstrass). *For every  $f \in C([a, b])$  and every  $\varepsilon > 0$  there exists a polynomial  $p$  such that*

$$\sup_{x \in [a, b]} |f(x) - p(x)| < \varepsilon.$$

*Equivalently, polynomials are dense in  $C([a, b])$  with respect to  $\|\cdot\|_\infty$ .*



**Constructive proof on  $[0, 1]$  (Bernstein polynomials).** For  $n \in \mathbb{N}$  and  $x \in [0, 1]$ , define

$$B_n(f, x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

Then  $B_n(f, \cdot)$  is a polynomial. Using the probabilistic identity

$$B_n(f, x) = \mathbb{E}\left[f\left(\frac{S_n}{n}\right)\right], \quad S_n \sim \text{Bin}(n, x),$$

together with  $\mathbb{E}[S_n/n] = x$  and  $\text{Var}(S_n/n) = x(1-x)/n$ , and writing  $\omega_f(\delta) = \sup_{|u-v| \leq \delta} |f(u) - f(v)|$  (modulus of continuity), one obtains

$$|B_n(f, x) - f(x)| \leq \omega_f\left(\sqrt{\frac{x(1-x)}{n}}\right) \leq \omega_f\left(\frac{1}{2\sqrt{n}}\right).$$

Hence  $\|B_n(f) - f\|_\infty \rightarrow 0$  as  $n \rightarrow \infty$ . For a general interval  $[a, b]$ , compose with the affine map  $[a, b] \leftrightarrow [0, 1]$ .

**Stone–Weierstrass (context).** More generally, if  $K$  is compact and  $\mathcal{A} \subset C(K)$  is a subalgebra that separates points and contains constants, then  $\mathcal{A}$  is dense in  $C(K)$ . Taking  $K = [a, b]$  and  $\mathcal{A}$  the algebra of polynomials yields Weierstrass.

**Corollary (pointwise approximation on  $\mathbb{R}$ ).** If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is continuous, then there exist polynomials  $p_n$  with  $\sup_{|x| \leq n} |f(x) - p_n(x)| < 1/n$ . For any fixed  $x$ , this implies  $p_n(x) \rightarrow f(x)$  as  $n \rightarrow \infty$ .

## Bernstein operator: direct uniform convergence for $f(x) = x^3$

**Definition.** For  $f \in C[0, 1]$ , the Bernstein polynomial is

$$(B_n f)(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}, \quad x \in [0, 1].$$

Equivalently, if  $S_n \sim \text{Bin}(n, x)$ , then  $(B_n f)(x) = \mathbb{E}[f(S_n/n)]$ .

**Computation for  $f(x) = x^3$ .** Using the factorial-moment identity  $S^3 = (S)_3 + 3(S)_2 + (S)_1$  and  $\mathbb{E}[(S_n)_r] = n^r x^r$  with  $n^r = n(n-1) \cdots (n-r+1)$ ,

$$B_n(x^3)(x) = \frac{n^3 x^3 + 3n^2 x^2 + nx}{n^3} = x^3 + \frac{3x^2 - 3x^3}{n} + \frac{x - 3x^2 + 2x^3}{n^2}.$$

Hence

$$B_n(x^3)(x) - x^3 = \frac{3x^2(1-x)}{n} + \frac{x(1-x)(1-2x)}{n^2}.$$

**Uniform bound.** On  $[0, 1]$ ,  $0 \leq x^2(1-x) \leq \frac{4}{27}$  and  $|x(1-x)(1-2x)| \leq \frac{1}{4}$ . Therefore

$$\sup_{x \in [0, 1]} |B_n(x^3)(x) - x^3| \leq \frac{4}{9n} + \frac{1}{4n^2} \xrightarrow{n \rightarrow \infty} 0.$$

Thus  $B_n f \rightarrow f$  uniformly on  $[0, 1]$  for  $f(x) = x^3$ . □

**Direct computation from the Bernstein formula (no expectation).** For  $t \in [0, 1]$ ,

$$B_n(x^3)(t) = \sum_{k=0}^n \left(\frac{k}{n}\right)^3 \binom{n}{k} t^k (1-t)^{n-k}.$$

Using  $k^3 = (k)_3 + 3(k)_2 + (k)_1$  and the identities

$$k \binom{n}{k} = n \binom{n-1}{k-1}, \quad k(k-1) \binom{n}{k} = n(n-1) \binom{n-2}{k-2}, \quad k(k-1)(k-2) \binom{n}{k} = n(n-1)(n-2) \binom{n-3}{k-3},$$

we obtain

$$\sum_{k=0}^n k^3 \binom{n}{k} t^k (1-t)^{n-k} = n(n-1)(n-2) t^3 + 3n(n-1) t^2 + n t.$$

Dividing by  $n^3$  yields

$$B_n(x^3)(t) = t^3 + \frac{3t^2 - 3t^3}{n} + \frac{t - 3t^2 + 2t^3}{n^2},$$

so

$$B_n(x^3)(t) - t^3 = \frac{3t^2(1-t)}{n} + \frac{t(1-t)(1-2t)}{n^2}.$$

Since  $0 \leq t^2(1-t) \leq \frac{4}{27}$  and  $|t(1-t)(1-2t)| \leq \frac{1}{4}$  on  $[0, 1]$ ,

$$\sup_{t \in [0,1]} |B_n(x^3)(t) - t^3| \leq \frac{4}{9n} + \frac{1}{4n^2} \rightarrow 0,$$

hence  $B_n(x^3) \rightarrow x^3$  uniformly on  $[0, 1]$ .

### Chebyshev polynomials (first kind)

They are defined by  $T_n(\cos \theta) = \cos(n\theta)$  and satisfy  $T_0(x) = 1$ ,  $T_1(x) = x$ , and  $T_{n+1}(x) = 2x T_n(x) - T_{n-1}(x)$ .

The first five are

$$\begin{aligned} T_0(x) &= 1, \\ T_1(x) &= x, \\ T_2(x) &= 2x^2 - 1, \\ T_3(x) &= 4x^3 - 3x, \\ T_4(x) &= 8x^4 - 8x^2 + 1. \end{aligned}$$

(Optionally,  $T_5(x) = 16x^5 - 20x^3 + 5x$ .)

### Legendre polynomials: Gram–Schmidt, endpoint vanishing, and Rodrigues-type $P_n$

**Inner product.**  $\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx.$

#### (i) First four Legendre polynomials via Gram–Schmidt

$$p_0(x) = 1, \quad p_1(x) = x, \quad p_2(x) = \frac{1}{2}(3x^2 - 1), \quad p_3(x) = \frac{1}{2}(5x^3 - 3x).$$

(Outline: orthogonalize  $1, x, x^2, x^3$ . Use  $\langle x, 1 \rangle = 0$ ,  $x^2 - (\langle x^2, 1 \rangle / \langle 1, 1 \rangle) = x^2 - \frac{1}{3}$ , and  $x^3 - (\langle x^3, x \rangle / \langle x, x \rangle)x = x^3 - \frac{3}{5}x$ , then scale to the standard normalization  $p_n(1) = 1$ .)

## (ii) Endpoint vanishing and orthogonality

For  $\phi_n(x) = (1-x)^n(1+x)^n = (1-x^2)^n$ ,  $x = \pm 1$  is a zero of multiplicity  $n$ , hence

$$\frac{d^m}{dx^m} \phi_n(\pm 1) = 0 \quad \text{for all } m < n.$$

Define  $P_n(x) := \frac{d^n}{dx^n} (1-x^2)^n$ . For  $m < n$ ,

$$\int_{-1}^1 P_n(x) P_m(x) dx = \int \phi_n^{(n)} \phi_m^{(m)} = (-1)^n \int \phi_n \phi_n^{(m+n)} = 0,$$

where we integrated by parts  $n$  times; all boundary terms vanish by the endpoint-vanishing above, and  $\phi_m^{(m+n)} \equiv 0$  since  $\deg \phi_m = 2m < m+n$ . Hence the  $P_n$  form an orthogonal family of degree  $n$  and therefore are scalar multiples of the Legendre polynomials  $p_n$ .

## (iii) Computing $P_n$ for $n = 0, 1, 2, 3$

$$\begin{aligned} P_0(x) &= 1, \\ P_1(x) &= -2x = (-2)p_1(x), \\ P_2(x) &= -4 + 12x^2 = 8p_2(x), \\ P_3(x) &= 72x - 120x^3 = (-48)p_3(x). \end{aligned}$$

Thus  $P_n = (-1)^n 2^n n! p_n$  for  $n = 0, 1, 2, 3$ , confirming the conclusion in (ii).

## Moments determine a continuous function

**Claim.** If  $f \in C[0, 1]$  and  $\int_0^1 f(x) x^n dx = 0$  for all  $n = 0, 1, 2, \dots$ , then  $f \equiv 0$ . If instead  $\int_0^1 f(x) x^n dx = 0$  holds for all  $n \geq 1$ , the same conclusion still holds.

**Proof (case  $n \geq 0$ ).** Define  $L : C[0, 1] \rightarrow \mathbb{R}$  by  $L(g) = \int_0^1 f(x)g(x) dx$ . Then  $L$  is continuous and  $L(p) = 0$  for every polynomial  $p$ . By the Weierstrass theorem, polynomials are dense in  $C[0, 1]$ , hence  $L \equiv 0$ . Taking  $g = f$  gives  $\int_0^1 f(x)^2 dx = 0$ , so  $f \equiv 0$ .

**Proof (case  $n \geq 1$  only).** Again let  $L(g) = \int_0^1 f(x)g(x) dx$ . If  $p$  is a polynomial with  $p(0) = 0$ , then  $p(x) = \sum_{k=1}^m a_k x^k$ , so  $L(p) = 0$  by hypothesis. By density of such polynomials in  $\{g \in C[0, 1] : g(0) = 0\}$ , we obtain  $L(g) = 0$  for every continuous  $g$  with  $g(0) = 0$ . Therefore, for arbitrary  $\varphi \in C[0, 1]$ ,

$$L(\varphi) = L(\varphi - \varphi(0)) + \varphi(0)L(1) = c\varphi(0), \quad c := \int_0^1 f.$$

Let  $\varphi_\varepsilon$  be a continuous function with  $\varphi_\varepsilon(0) = 1$  and  $\text{supp } \varphi_\varepsilon \subset [0, \varepsilon]$ . Then

$$c = L(\varphi_\varepsilon) = \int_0^1 f(x)\varphi_\varepsilon(x) dx \quad \text{and} \quad |L(\varphi_\varepsilon)| \leq \|f\|_\infty \varepsilon \rightarrow 0,$$

hence  $c = 0$ . Thus  $L(\varphi) = 0$  for all  $\varphi \in C[0, 1]$ , and taking  $\varphi = f$  yields  $\int_0^1 f(x)^2 dx = 0$ . By continuity,  $f \equiv 0$ .  $\square$

**Remark.** Without continuity, the second statement is false: the point mass  $\delta_0$  has  $\int_0^1 x^n d\delta_0 = 0$  for all  $n \geq 1$  but is not the zero function.

## Orthogonalization of $1, x, x^2, \dots$ on $[-1, 1]$ (Legendre via Gram–Schmidt)

**Inner product.**  $\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$ . Useful integrals:  $\int_{-1}^1 x^m dx = 0$  for  $m$  odd, and  $= 2/(m+1)$  for  $m$  even.

**Results** (scaled so  $p_n(1) = 1$ ).

$$\begin{aligned} p_0(x) &= 1, \\ p_1(x) &= x, \\ p_2(x) &= \frac{1}{2}(3x^2 - 1), \\ p_3(x) &= \frac{1}{2}(5x^3 - 3x), \\ p_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3). \end{aligned}$$

**Sketch of Gram–Schmidt steps.**

$$\begin{aligned} u_0 &= 1 \Rightarrow p_0 = 1, \\ u_1 &= x - \frac{\langle x, 1 \rangle}{\langle 1, 1 \rangle} 1 = x \Rightarrow p_1 = x, \\ u_2 &= x^2 - \frac{\langle x^2, 1 \rangle}{\langle 1, 1 \rangle} 1 = x^2 - \frac{1}{3} \Rightarrow p_2 = \frac{3}{2}\left(x^2 - \frac{1}{3}\right), \\ u_3 &= x^3 - \frac{\langle x^3, x \rangle}{\langle x, x \rangle} x = x^3 - \frac{3}{5}x \Rightarrow p_3 = \frac{5}{2}\left(x^3 - \frac{3}{5}x\right), \\ u_4 &= x^4 - \frac{\langle x^4, 1 \rangle}{\langle 1, 1 \rangle} 1 - \frac{\langle x^4, p_2 \rangle}{\langle p_2, p_2 \rangle} p_2 = x^4 - \frac{6}{7}x^2 + \frac{3}{35}, \\ p_4 &= \frac{35}{8} u_4 = \frac{1}{8}(35x^4 - 30x^2 + 3). \end{aligned}$$

**Orthonormal scaling.**  $\int_{-1}^1 p_n^2 = \frac{2}{2n+1}$ , so  $\phi_n(x) = \sqrt{\frac{2n+1}{2}} p_n(x)$  is orthonormal.

## Orthogonality of Chebyshev polynomials and the weight

Let  $T_n$  be the Chebyshev polynomials of the first kind, defined by  $T_n(\cos \theta) = \cos(n\theta)$ . Set

$$w(x) = \frac{1}{\sqrt{1-x^2}}, \quad x \in (-1, 1).$$

Then for all  $m, n \geq 0$ ,

$$\int_{-1}^1 T_m(x) T_n(x) w(x) dx = \int_{-1}^1 T_m(x) T_n(x) \frac{dx}{\sqrt{1-x^2}} = \int_0^\pi \cos(m\theta) \cos(n\theta) d\theta,$$

where we used the substitution  $x = \cos \theta$  (so  $dx = -\sin \theta d\theta$  and  $dx/\sqrt{1-x^2} = -d\theta$ ). By the standard cosine orthogonality on  $[0, \pi]$ ,

$$\int_0^\pi \cos(m\theta) \cos(n\theta) d\theta = \begin{cases} 0, & m \neq n, \\ \pi, & m = n = 0, \\ \pi/2, & m = n \geq 1. \end{cases}$$

Therefore  $\{T_n\}_{n \geq 0}$  is an orthogonal system on  $[-1, 1]$  with respect to  $w(x) = (1 - x^2)^{-1/2}$ . An orthonormal system is given by

$$\phi_0(x) = \frac{1}{\sqrt{\pi}}, \quad \phi_n(x) = \sqrt{\frac{2}{\pi}} T_n(x) \quad (n \geq 1).$$

## Quadrature with nodes and weights; exactness and Gauss rules

Fix an interval  $[a, b]$ . For  $n \in \mathbb{N}$ , let nodes  $x_k^{(n)} \in [a, b]$  and weights  $A_k^{(n)} \geq 0$  be given ( $k = 1, \dots, n$ ). Define

$$Q_n(P) := \sum_{k=1}^n A_k^{(n)} P(x_k^{(n)}), \quad \varepsilon_n(P) := \left| \int_a^b P(x) dx - Q_n(P) \right|.$$

**Exactness and moment conditions.** The rule is exact for all polynomials of degree  $\leq m$  if and only if

$$\sum_{k=1}^n A_k^{(n)} (x_k^{(n)})^j = \int_a^b x^j dx \quad (j = 0, 1, \dots, m),$$

i.e. the first  $m+1$  moments are matched.

**Sharp upper bound.** No  $n$ -node rule can be exact beyond degree  $2n - 1$ . Indeed, if it were exact for degree  $2n$ , then with  $\pi(x) = \prod_{k=1}^n (x - x_k^{(n)})^2 \geq 0$  we would get  $0 = Q_n(\pi) = \sum_k A_k^{(n)} \pi(x_k^{(n)}) = \int_a^b \pi(x) dx > 0$ , a contradiction.

**Gauss–Legendre rule (achieves  $2n-1$ ).** Let  $P_n$  be the degree- $n$  Legendre polynomial on  $[-1, 1]$  and let  $\xi_1, \dots, \xi_n$  be its zeros. Set nodes on  $[a, b]$  by  $x_k^{(n)} = \frac{a+b}{2} + \frac{b-a}{2} \xi_k$  and weights by

$$A_k^{(n)} = \frac{b-a}{2} \omega_k, \quad \omega_k = \frac{2}{(1 - \xi_k^2) [P_n'(\xi_k)]^2} (> 0).$$

Then

$$\sum_{k=1}^n A_k^{(n)} P(x_k^{(n)}) = \int_a^b P(x) dx \quad \text{for all polynomials } \deg P \leq 2n - 1.$$

*Sketch:* write  $p = qP_n + r$  with  $\deg q, \deg r \leq n-1$ ; orthogonality gives  $\int qP_n = 0$  and  $P_n(\xi_k) = 0$  gives  $p(\xi_k) = r(\xi_k)$ ; the quadrature integrates  $r$  exactly.

**Examples on  $[-1, 1]$ .**

- $n = 1$ : node 0, weight 2; exact for  $\deg \leq 1$ .
- $n = 2$ : nodes  $\pm 1/\sqrt{3}$ , weights 1, 1; exact for  $\deg \leq 3$ .
- $n = 3$ : nodes 0,  $\pm\sqrt{3/5}$ ; weights 8/9, 5/9, 5/9; exact for  $\deg \leq 5$ .

Mapping to  $[a, b]$  uses  $x = \frac{a+b}{2} + \frac{b-a}{2} \xi$  and multiplies all weights by  $\frac{b-a}{2}$ .

## Function Approximation vs. Quadrature (on $[-1, 1]$ )

### A. Polynomial approximation of a continuous function

**Goal.** Given  $f \in C([-1, 1])$ , choose a polynomial  $p_n$  so that

$$\|f - p_n\|_\infty = \sup_{x \in [-1, 1]} |f(x) - p_n(x)|$$

is small (ideally, minimal). This targets the *entire function* on the whole interval.

**Tools.** Weierstrass approximation (existence of good polynomials), Chebyshev (minimax/near-best choices), Jackson-type estimates (rates in terms of smoothness).

## B. Numerical quadrature (this task)

**Goal.** Approximate the *single number*

$$I(f) = \int_{-1}^1 f(x) dx$$

by a weighted sum of *finitely many samples*:

$$Q_n(f) = \sum_{k=1}^n A_k f(x_k), \quad x_k \in [-1, 1], \quad A_k > 0.$$

We design  $x_k, A_k$  so that  $Q_n$  is *exact* for all polynomials of degree  $\leq m$ :

$$Q_n(p) = \int_{-1}^1 p(x) dx \quad \text{for all } \deg p \leq m.$$

Any  $n$ -node rule has  $m \leq 2n - 1$  (sharp). *Gauss–Legendre* quadrature (zeros of the Legendre polynomial  $P_n$  as nodes, specific positive weights) achieves  $m = 2n - 1$ .

## C. Same actors, different roles

Orthogonal polynomials appear in both settings:

- In *approximation*, expanding  $f$  in Chebyshev/Legendre series and truncating gives small  $\|f - p_n\|_\infty$ .
- In *quadrature*, Gauss nodes are the zeros of an orthogonal polynomial, yielding exactness up to degree  $2n - 1$ .

## D. Tiny example

For  $f(x) = x^4$  on  $[-1, 1]$ :

- A degree-1 polynomial  $p(x) = ax + b$  cannot uniformly approximate  $x^4$  well; the optimal error is  $O(1)$ .
- The 3-point Gauss rule (degree of exactness 5) integrates  $x^4$  *exactly* using just 3 samples, even though it does not approximate the whole function.

## E. Side-by-side comparison

| Aspect     | Approximation                  | Quadrature                          |
|------------|--------------------------------|-------------------------------------|
| Target     | $f(\cdot)$ on $[-1, 1]$        | $\int_{-1}^1 f$ (a scalar)          |
| Output     | Polynomial $p_n$               | Nodes $x_k$ , weights $A_k$         |
| Error      | $\ f - p_n\ _\infty$ (uniform) | $ f - \sum A_k f(x_k) $             |
| Optimality | Minimax/Chebyshev, etc.        | Degree of exactness $m \leq 2n - 1$ |
| Data used  | Potentially dense info on $f$  | Only $n$ samples $f(x_k)$           |
| Key tools  | Weierstrass, Chebyshev series  | Orthogonal polys, Gauss nodes       |

## No $n$ -node quadrature can be exact beyond degree $2n - 1$

Let  $\alpha_1, \dots, \alpha_n \in [-1, 1]$  be distinct nodes and  $A_1, \dots, A_n \in \mathbb{R}$  weights. Consider the quadrature  $Q_n(p) = \sum_{j=1}^n A_j p(\alpha_j)$ .

**Theorem 4.** *For any choice of nodes and weights, the degree of exactness of  $Q_n$  is at most  $2n - 1$ . In particular, there do not exist  $n$  distinct nodes and real weights such that*

$$\int_{-1}^1 p(x) dx = \sum_{j=1}^n A_j p(\alpha_j) \quad \text{for all } \deg p \leq 2n.$$

*Proof.* Define  $\pi(x) = \prod_{j=1}^n (x - \alpha_j)^2$ . Then  $\deg \pi = 2n$ , and  $\pi(\alpha_j) = 0$  for all  $j$ , so  $Q_n(\pi) = 0$ .

Since  $\pi \geq 0$  and  $\pi \not\equiv 0$ , we have  $\int_{-1}^1 \pi(x) dx > 0$ . If the rule were exact for degree  $2n$ , we would have  $Q_n(\pi) = \int_{-1}^1 \pi$ , a contradiction. Hence the degree of exactness is  $\leq 2n - 1$ .  $\square$

**Optimality.** The Gauss–Legendre rule, with nodes the zeros of the Legendre polynomial  $P_n$  and the usual positive weights, is exact for all  $\deg p \leq 2n - 1$ , so the bound is sharp.

**Checks.** For  $n = 1$ , exactness for  $1, x, x^2$  forces  $A_1 = 2$ ,  $\alpha_1 = 0$ , and  $2\alpha_1^2 = 2/3$ , impossible. For  $n = 2$  with symmetric nodes  $\pm c$  and equal weights  $w$ , exactness for  $1, x^2, x^4$  yields  $w = 1$ ,  $c^2 = 1/3$ , and  $c^4 = 1/5$ , a contradiction.

## What an $n$ -point quadrature rule means

**Definition.** An  $n$ -point quadrature rule on  $[a, b]$  is specified by *nodes*  $x_1, \dots, x_n \in [a, b]$  and *weights*  $A_1, \dots, A_n$  and approximates

$$\int_a^b f(x) dx \approx Q_n(f) := \sum_{k=1}^n A_k f(x_k).$$

Thus the rule requires exactly  $n$  function evaluations  $f(x_k)$ .

**Degree of exactness.** The rule is *exact of degree  $m$*  if  $Q_n(p) = \int_a^b p(x) dx$  for every polynomial  $p$  with  $\deg p \leq m$ . With only function values (no derivatives), any  $n$ -point rule satisfies  $m \leq 2n - 1$  (sharp). Gauss–Legendre quadrature attains  $m = 2n - 1$ .

**Closed vs. open rules.** Nodes may include endpoints (*closed* rules, e.g. trapezoid, Simpson) or avoid them (*open* rules, e.g. midpoint). In either case there are  $n$  nodes and  $n$  evaluations.

**Composite/adaptive use.** To integrate on  $[a, b]$  one often partitions the interval and applies the  $n$ -point rule on each subinterval; the overall method then uses many points (roughly  $n$  times the number of panels), but per panel it remains an  $n$ -point quadrature.

**Examples on  $[-1, 1]$ .**

- 2-point Gauss: nodes  $\pm 1/\sqrt{3}$ , weights 1, 1; exact for  $\deg \leq 3$ .
- Simpson's rule (3 points): nodes  $-1, 0, 1$ , weights  $\frac{1}{3}(1, 4, 1)$ ; exact for  $\deg \leq 3$ .

## Examples of $n$ -point quadrature rules and what you evaluate

**Definition.** An  $n$ -point rule on  $[a, b]$  has nodes  $x_1, \dots, x_n \in [a, b]$  and weights  $A_1, \dots, A_n$  and computes

$$\int_a^b f(x) dx \approx Q_n(f) := \sum_{k=1}^n A_k f(x_k),$$

so you evaluate  $f$  at exactly  $n$  points  $x_k$ .

**Standard rules (nodes, weights, degree of exactness).** On  $[a, b]$  we have:

| Rule                       | Nodes $x_k$                                                        | Weights $A_k$                                                            | Exact for deg $\leq$ |
|----------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------|
| Midpoint ( $n = 1$ )       | $\frac{a+b}{2}$                                                    | $b - a$                                                                  | 1                    |
| Trapezoid ( $n = 2$ )      | $a, b$                                                             | $\frac{b-a}{2}, \frac{b-a}{2}$                                           | 1                    |
| Simpson ( $n = 3$ )        | $a, \frac{a+b}{2}, b$                                              | $\frac{b-a}{6} \cdot (1, 4, 1)$                                          | 3                    |
| Gauss-Legendre ( $n = 2$ ) | $\frac{a+b}{2} \pm \frac{b-a}{2\sqrt{3}}$                          | $\frac{b-a}{2}, \frac{b-a}{2}$                                           | 3                    |
| Gauss-Legendre ( $n = 3$ ) | $\frac{a+b}{2}, \frac{a+b}{2} \pm \frac{b-a}{2}\sqrt{\frac{3}{5}}$ | $\frac{b-a}{2} \cdot \left(\frac{8}{9}, \frac{5}{9}, \frac{5}{9}\right)$ | 5                    |

**What you actually compute.** For example, with the 3-point Gauss rule on  $[a, b]$ :

$$Q_3(f) = \frac{b-a}{2} \left[ \frac{8}{9} f\left(\frac{a+b}{2}\right) + \frac{5}{9} f\left(\frac{a+b}{2} - \frac{b-a}{2}\sqrt{\frac{3}{5}}\right) + \frac{5}{9} f\left(\frac{a+b}{2} + \frac{b-a}{2}\sqrt{\frac{3}{5}}\right) \right],$$

which uses *exactly three* evaluations of  $f$ .

**Sanity check on  $[-1, 1]$  for  $f(x) = x^4$  (true integral =  $2/5$ ).**

$$\text{Simpson: } \frac{2}{6} [f(-1) + 4f(0) + f(1)] = \frac{1}{3}(1 + 0 + 1) = \frac{2}{3},$$

$$\text{Gauss 2-point: } f(-1/\sqrt{3}) + f(1/\sqrt{3}) = 2 \cdot \left(\frac{1}{\sqrt{3}}\right)^4 = \frac{2}{9},$$

$$\text{Gauss 3-point: } \frac{8}{9} f(0) + \frac{5}{9} f(\pm\sqrt{3/5}) = \frac{10}{9} \left(\frac{3}{5}\right)^2 = \frac{2}{5}.$$

Only the 3-point Gauss rule (degree 5) integrates the quartic exactly.

## Approximating integrals by exactness on polynomials; Legendre background

**Quadrature and degree of exactness.** An  $n$ -point rule on  $[-1, 1]$  has nodes  $x_1, \dots, x_n$  and weights  $A_1, \dots, A_n$  and computes

$$Q_n(f) = \sum_{k=1}^n A_k f(x_k).$$

It is *exact of degree  $m$*  if  $Q_n(p) = \int_{-1}^1 p(x) dx$  for every polynomial  $p$  with  $\deg p \leq m$ . With  $n$  nodes (no derivatives) one always has  $m \leq 2n - 1$  (sharp).

**Legendre polynomials.** They form an orthogonal basis on  $[-1, 1]$  (weight 1):

$$\int_{-1}^1 P_n(x) P_m(x) dx = \begin{cases} \frac{2}{2n+1}, & m = n, \\ 0, & m \neq n. \end{cases}$$

Rodrigues:  $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$ . Recurrence:  $(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$ .



**Gauss–Legendre quadrature (achieves  $2n-1$ ).** Let  $\xi_1, \dots, \xi_n$  be the zeros of  $P_n$ . Set

$$\omega_k = \frac{2}{(1 - \xi_k^2) [P'_n(\xi_k)]^2} \quad (\text{all } \omega_k > 0), \quad Q_n(f) = \sum_{k=1}^n \omega_k f(\xi_k).$$

Then  $Q_n(p) = \int_{-1}^1 p(x) dx$  for every  $\deg p \leq 2n-1$ . *Sketch:* write  $p = qP_n + r$  with  $\deg q, \deg r \leq n-1$ ; use  $\int qP_n = 0$  and  $P_n(\xi_k) = 0$ , and the fact that  $Q_n$  integrates any degree  $\leq n-1$  polynomial exactly.

**Examples on  $[-1, 1]$ .**

- $n = 2$ : nodes  $\pm 1/\sqrt{3}$ , weights 1, 1; exact for  $\deg \leq 3$ .
- $n = 3$ : nodes 0,  $\pm\sqrt{3/5}$ , weights 8/9, 5/9, 5/9; exact for  $\deg \leq 5$ .

**Impossibility beyond  $2n-1$ .** For any nodes  $\{\alpha_j\}$ , the polynomial  $\pi(x) = \prod_{j=1}^n (x - \alpha_j)^2$  has degree  $2n$ , satisfies  $Q_n(\pi) = 0$ , but  $\int_{-1}^1 \pi(x) dx > 0$ . Hence no  $n$ -point rule can be exact for all  $\deg 2n$  polynomials.

## Building a 10-point quadrature for $[a, b]$

**Goal.** Approximate  $I(f) = \int_a^b f(x) dx$  from  $n = 10$  samples of  $f$ .

### A. Choosing the nodes (Gauss–Legendre, optimal)

Let  $(\xi_i, \omega_i)$  be the 10-point Gauss–Legendre nodes/weights on  $[-1, 1]$ :

$$\begin{aligned} \xi_{1,10} &= \pm 0.9739065285171717, & \omega_{1,10} &= 0.0666713443086881, \\ \xi_{2,9} &= \pm 0.8650633666889845, & \omega_{2,9} &= 0.1494513491505806, \\ \xi_{3,8} &= \pm 0.6794095682990244, & \omega_{3,8} &= 0.2190863625159820, \\ \xi_{4,7} &= \pm 0.4333953941292472, & \omega_{4,7} &= 0.2692667193099963, \\ \xi_{5,6} &= \pm 0.1488743389816312, & \omega_{5,6} &= 0.2955242247147529. \end{aligned}$$

Map to  $[a, b]$  by

$$x_i = \frac{a+b}{2} + \frac{b-a}{2} \xi_i, \quad A_i = \frac{b-a}{2} \omega_i,$$

and compute

$$Q_{10}(f) = \sum_{i=1}^{10} A_i f(x_i).$$

This rule is exact for all polynomials with  $\deg \leq 19$ .

### B. Nodes are given (interpolatory quadrature)

Let distinct  $x_1, \dots, x_{10} \in [a, b]$  be given and define the Lagrange basis

$$\ell_k(x) = \prod_{\substack{j=1 \\ j \neq k}}^{10} \frac{x - x_j}{x_k - x_j}.$$

Set the weights by

$$A_k = \int_a^b \ell_k(x) dx, \quad k = 1, \dots, 10.$$

Then

$$Q_{10}(f) = \sum_{k=1}^{10} A_k f(x_k)$$

is exact for all polynomials of degree  $\leq 9$ . Equivalently,  $\{A_k\}$  solve the moment system

$$\sum_{k=1}^{10} A_k x_k^j = \frac{b^{j+1} - a^{j+1}}{j+1}, \quad j = 0, 1, \dots, 9.$$

**Remarks.** (1) Affine-scaling to  $[-1, 1]$  before forming  $\ell_k$  or the moment system improves conditioning.

(2) If the nodes are equally spaced (including endpoints), this yields the closed Newton–Cotes rule with 10 points, which is less stable than Gauss–Legendre.

## Chebyshev tail with nonnegative coefficients: equioscillation of the truncation error

Let  $T_j$  be the Chebyshev polynomials of the first kind and let  $\{\gamma_j\}_{j \geq 1}$  be nonnegative with  $\sum_{j=1}^{\infty} \gamma_j < \infty$ . Define

$$f(x) = \sum_{j=1}^{\infty} \gamma_j T_{3j}(x), \quad P_n(x) = \sum_{j=1}^n \gamma_j T_{3j}(x), \quad R_n(x) = f(x) - P_n(x) = \sum_{j=n+1}^{\infty} \gamma_j T_{3j}(x).$$

**(1) Continuity.** Since  $|T_{3j}(x)| \leq 1$  on  $[-1, 1]$  and  $\sum \gamma_j < \infty$ , the series for  $f$  converges absolutely and uniformly (M-test), so  $f \in C([-1, 1])$ .

**(2) Exact size of the tail.** For all  $x \in [-1, 1]$ ,  $|R_n(x)| \leq \sum_{j=n+1}^{\infty} \gamma_j =: E_n$ . At  $x = 1$ ,  $T_{3j}(1) = 1$ , hence  $|R_n(1)| = E_n$ . Therefore  $\|f - P_n\|_{\infty} = \|R_n\|_{\infty} = E_n$ .

**(3) Minimax property of  $P_n$ .** Let  $\mathcal{P}_{3n}$  be the polynomials of degree  $\leq 3n$ . For any  $Q \in \mathcal{P}_{3n}$ , write  $Q = \sum_{k=0}^{3n} a_k T_k$ . Then

$$f - Q = \sum_{j=1}^n (\gamma_j - a_{3j}) T_{3j} + \sum_{\substack{0 \leq k \leq 3n \\ k \not\equiv 0 \pmod{3}}} (-a_k) T_k + R_n.$$

Using  $\|\sum c_{\ell} T_{\ell}\|_{\infty} \leq \sum |c_{\ell}|$  gives

$$\|f - Q\|_{\infty} \geq \|R_n\|_{\infty} - \sum_{j=1}^n |\gamma_j - a_{3j}| - \sum_{\substack{k \leq 3n \\ k \not\equiv 0 \pmod{3}}} |a_k| \geq \|R_n\|_{\infty} = E_n,$$

with equality iff  $a_{3j} = \gamma_j$  for  $1 \leq j \leq n$  and  $a_k = 0$  for  $k \not\equiv 0 \pmod{3}$ , i.e. iff  $Q = P_n$ . Thus  $P_n$  is the minimax polynomial in  $\mathcal{P}_{3n}$  and the best error equals  $E_n$ .

**(4) Equioscillation points.** By the Chebyshev alternation theorem for best uniform approximation by  $\mathcal{P}_{3n}$ , there exist points

$$-1 \leq x_0 < x_1 < \dots < x_{3n+1} \leq 1$$

such that

$$f(x_k) - P_n(x_k) = (-1)^k \|f - P_n\|_{\infty} = (-1)^k \sum_{j=n+1}^{\infty} \gamma_j, \quad k = 0, 1, \dots, 3n+1.$$

## Using 10 points to approximate $\int_a^b f(x) dx$

**Gauss–Legendre (you may choose the points).** Let  $(\xi_i, \omega_i)_{i=1}^{10}$  be the 10-point Gauss–Legendre nodes/weights on  $[-1, 1]$ . Map to  $[a, b]$  by

$$x_i = \frac{a+b}{2} + \frac{b-a}{2} \xi_i, \quad A_i = \frac{b-a}{2} \omega_i.$$

Then the quadrature

$$Q_{10}(f) = \sum_{i=1}^{10} A_i f(x_i)$$

is exact for every polynomial  $p$  with  $\deg p \leq 19$  (optimal among all 10-point rules), and all  $A_i > 0$ .

**Interpolatory rule (points are pre-given).** If distinct  $x_1, \dots, x_{10} \in [a, b]$  are given, define the Lagrange basis

$$\ell_k(x) = \prod_{\substack{j=1 \\ j \neq k}}^{10} \frac{x - x_j}{x_k - x_j}, \quad A_k = \int_a^b \ell_k(x) dx.$$

Then

$$Q_{10}(f) = \sum_{k=1}^{10} A_k f(x_k)$$

is exact for all polynomials with  $\deg \leq 9$ . (Equivalently, the weights solve  $\sum_{k=1}^{10} A_k x_k^j = (b^{j+1} - a^{j+1})/(j+1)$  for  $j = 0, \dots, 9$ .)

## Problem 4

For each  $f \in C([-1, 1])$ , let  $\mathcal{P}_n$  denote the subspace of polynomials of degree at most  $n$ , and define the best uniform approximation error

$$E_n(f) := \inf_{p \in \mathcal{P}_n} \sup_{x \in [-1, 1]} |f(x) - p(x)|.$$

By the Weierstrass approximation theorem,  $E_n(f) \rightarrow 0$  for every  $f \in C([-1, 1])$ . Using the result of Question 3, construct a function  $f \in C([-1, 1])$  to show that the convergence  $E_n(f) \rightarrow 0$  can be *arbitrarily slow* in the following sense:

**Claim to prove.** For any decreasing sequence  $(\delta_n)_{n \geq 1}$  of nonnegative numbers with  $\delta_n \rightarrow 0$ , there exists  $f \in C([-1, 1])$  such that

$$E_n(f) \geq \delta_n \quad \text{for all } n = 1, 2, \dots$$

## Arbitrarily slow uniform polynomial approximation

For  $f \in C([-1, 1])$  let

$$E_n(f) := \inf_{p \in \mathcal{P}_n} \|f - p\|_\infty, \quad \mathcal{P}_n = \{p : \deg p \leq n\}.$$

By Weierstrass,  $E_n(f) \downarrow 0$ . We show the convergence can be arbitrarily slow.

**Lemma (from Problem 3).** If  $\gamma_j \geq 0$  with  $\sum_{j=1}^{\infty} \gamma_j < \infty$  and

$$f(x) = \sum_{j=1}^{\infty} \gamma_j T_{3j}(x), \quad P_n(x) = \sum_{j=1}^n \gamma_j T_{3j}(x),$$

then  $f \in C([-1, 1])$ ,  $P_n$  is the best uniform approximant of degree  $\leq 3n$ , and

$$E_{3n}(f) = \|f - P_n\|_{\infty} = \sum_{j=n+1}^{\infty} \gamma_j.$$

**Theorem (arbitrarily slow decay of  $E_n(f)$ ).** Given any decreasing sequence  $\delta_n \geq 0$  with  $\delta_n \rightarrow 0$ , there exists  $f \in C([-1, 1])$  such that  $E_n(f) \geq \delta_n$  for all  $n \geq 1$ .

*Proof.* Set  $s_n := \delta_{3n}$  and define  $\gamma_{n+1} := s_n - s_{n+1} \geq 0$ . Then  $\sum \gamma_j = s_0 - \lim s_n < \infty$  and hence  $f(x) := \sum_{j=1}^{\infty} \gamma_j T_{3j}(x) \in C([-1, 1])$ . By the lemma,

$$E_{3n}(f) = \sum_{j=n+1}^{\infty} \gamma_j = s_n = \delta_{3n}.$$

Since  $E_n(f)$  is nonincreasing in  $n$ , for any  $m \in \{3n, 3n+1, 3n+2\}$ ,

$$E_m(f) \geq E_{3n}(f) = \delta_{3n} \geq \delta_m,$$

because  $\delta_n$  is decreasing. Therefore  $E_n(f) \geq \delta_n$  for all  $n$ .  $\square$

**Examples.** Choosing  $\delta_n = 1/\log(n+e)$  (or  $1/\log \log(n+e^e)$ , etc.) yields a continuous  $f$  for which  $E_n(f)$  decreases as slowly as the prescribed sequence.

## Piecewise-linear (hat-function) interpolation on a uniform grid

**Definition.** For  $n \in \mathbb{N}$  and  $r \in \mathbb{Z}$ , define

$$\Delta_{n,r}(x) := \max\left\{0, 1 - n\left|x - \frac{r}{n}\right|\right\}, \quad x \in [-1, 1].$$

Each  $\Delta_{n,r}$  is the “hat” centered at  $r/n$  with height 1 and support  $[(r-1)/n, (r+1)/n]$ .

**Sketch.**  $\Delta_{n,r}$  is triangular: it is linear on  $[(r-1)/n, r/n]$  and on  $[r/n, (r+1)/n]$ , with

$$\Delta_{n,r}(r/n) = 1, \quad \Delta_{n,r}((r \pm 1)/n) = 0.$$

For fixed  $x$ , at most two hats are nonzero and they satisfy  $\sum_{m=-n}^n \Delta_{n,m}(x) = 1$ .

**Interpolator.** Given  $f : [-1, 1] \rightarrow \mathbb{R}$ , set

$$f_n(x) := \sum_{m=-n}^n f\left(\frac{m}{n}\right) \Delta_{n,m}(x).$$

**Lemma 2** (Piecewise linearity and interpolation). *For each  $n$ ,  $f_n$  is linear on each subinterval  $[r/n, (r+1)/n]$ , and  $f_n(r/n) = f(r/n)$  for  $r = -n, \dots, n$ .*

*Proof.* On  $[r/n, (r+1)/n]$  only  $\Delta_{n,r}$  and  $\Delta_{n,r+1}$  are nonzero, both linear, and  $\Delta_{n,r}(x) + \Delta_{n,r+1}(x) = 1$ . Hence

$$f_n(x) = \Delta_{n,r}(x) f(r/n) + \Delta_{n,r+1}(x) f((r+1)/n)$$

is linear there. At  $x = r/n$ ,  $\Delta_{n,r}(r/n) = 1$  and all other hats vanish, so  $f_n(r/n) = f(r/n)$ .  $\square$

**Theorem 5** (Uniform convergence for continuous data). *If  $f \in C([-1, 1])$ , then  $\|f_n - f\|_\infty \rightarrow 0$  as  $n \rightarrow \infty$ .*

*Proof.* Since  $f$  is uniformly continuous, for any  $\varepsilon > 0$  choose  $\delta > 0$  so that  $|f(x) - f(y)| < \varepsilon$  whenever  $|x - y| < \delta$ . Take  $n$  with  $1/n < \delta$ . For  $x \in [r/n, (r+1)/n]$ ,

$$\begin{aligned} |f_n(x) - f(x)| &= |\Delta_{n,r}(x)(f(r/n) - f(x)) + \Delta_{n,r+1}(x)(f((r+1)/n) - f(x))| \\ &\leq \Delta_{n,r}(x) |f(r/n) - f(x)| + \Delta_{n,r+1}(x) |f((r+1)/n) - f(x)| < \varepsilon, \end{aligned}$$

since both  $|x - r/n|$  and  $|x - (r+1)/n|$  are  $\leq 1/n < \delta$ . Taking the supremum over  $x$  gives  $\|f_n - f\|_\infty < \varepsilon$ .  $\square$

**Rate under Lipschitz continuity (optional).** If  $f$  is Lipschitz with constant  $L$ , then for all  $x$ ,

$$|f_n(x) - f(x)| \leq L \max \{|x - r/n|, |x - (r+1)/n|\} \leq \frac{L}{n},$$

hence  $\|f_n - f\|_\infty \leq L/n$ .

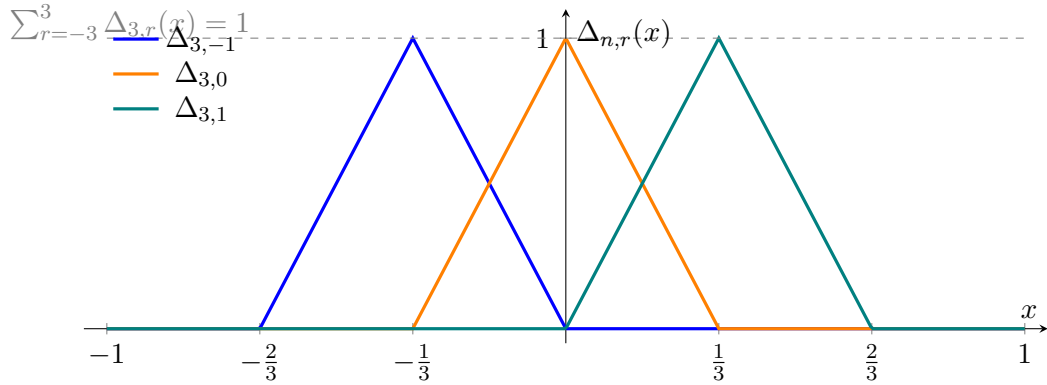


Figure 10: Three neighboring hat functions for  $n = 3$ . Each peak is at  $x = r/n$ ; support width is  $2/n$ .

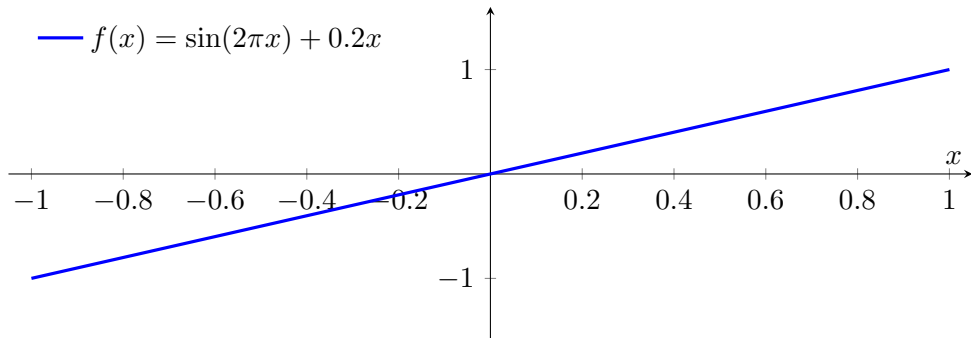


Figure 11: Piecewise-linear interpolant  $f_n$  on a uniform grid with  $n = 6$ .

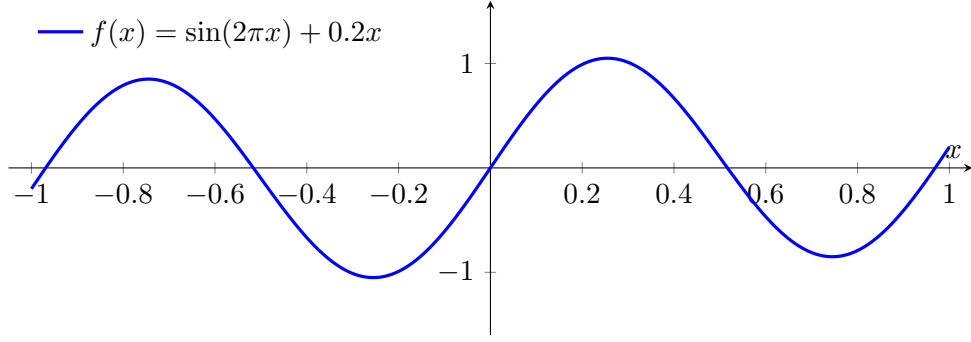


Figure 12: Target function  $f(x) = \sin(2\pi x) + 0.2x$ .

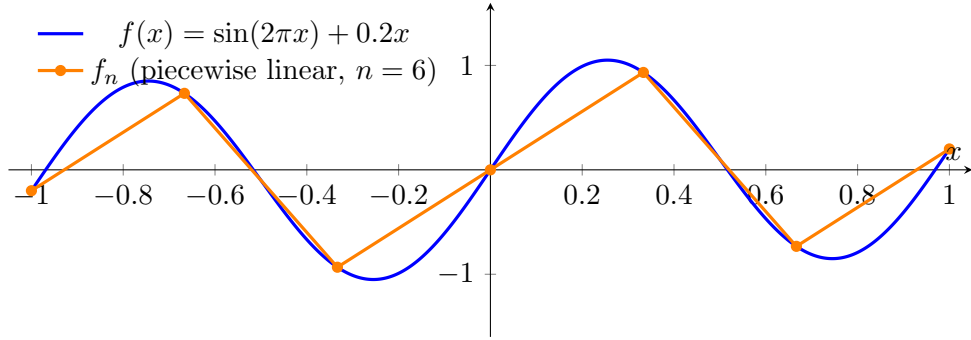


Figure 13: Target function  $f(x)$  and its piecewise-linear interpolant  $f_n$  for  $n = 6$ .

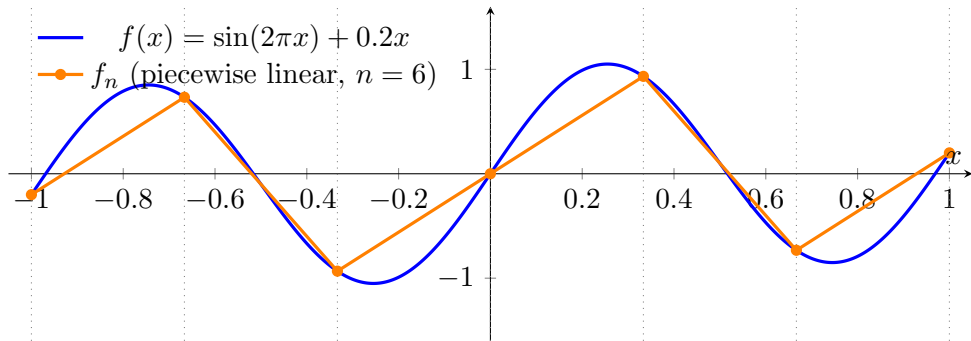


Figure 14: Target function  $f(x)$  and piecewise-linear interpolant  $f_n$  for  $n = 6$ , with vertical grid lines at nodes.

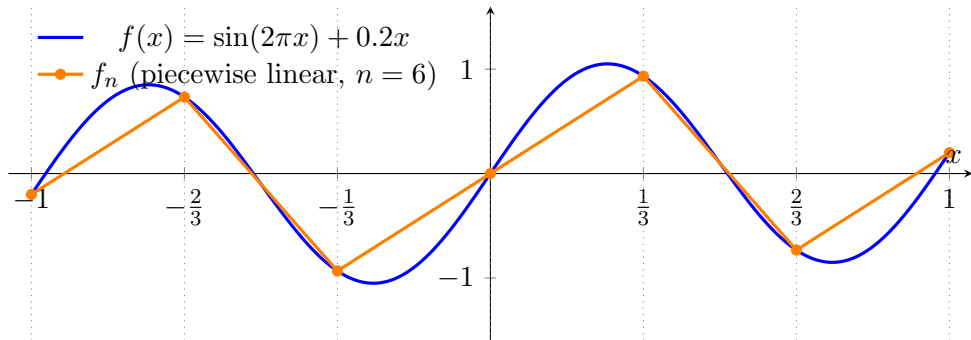


Figure 15: Comparison of  $f(x)$  and piecewise-linear interpolant  $f_n$  on a uniform grid with  $n = 6$ , with vertical grid lines at nodes.

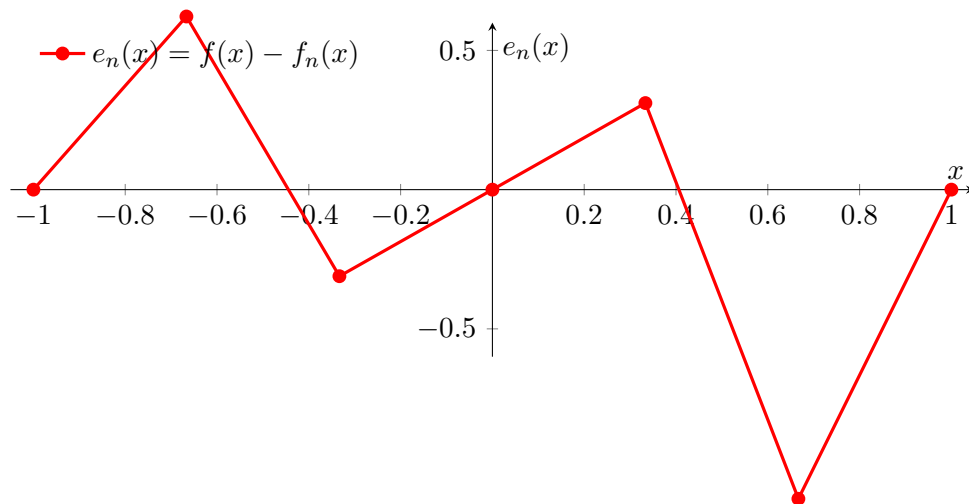


Figure 16: Interpolation error  $e_n(x)$  at nodes and midpoints for  $n = 6$ .

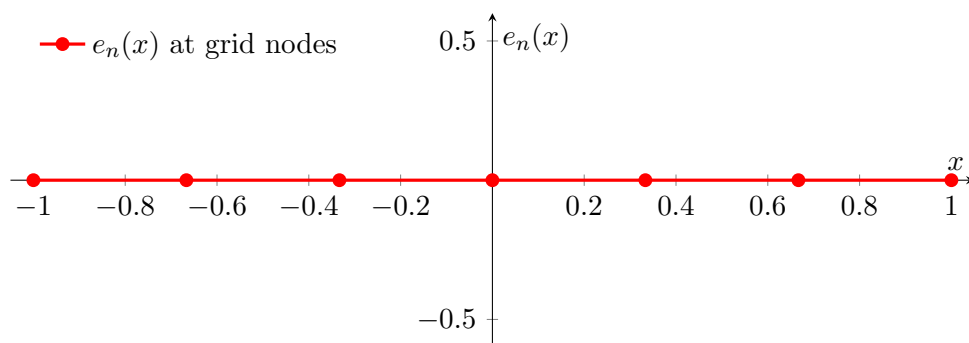


Figure 17: Interpolation error  $e_n(x) = f(x) - f_n(x)$  at grid nodes ( $n = 6$ ).

**Exercise 3. Schauder basis for  $C([-1, 1])$ .** Show that there exists a sequence  $(\phi_n)_{n \geq 0} \subset C([-1, 1])$  such that for every  $f \in C([-1, 1])$  there exist unique scalars  $(a_n)_{n \geq 0}$  with the series

$$\sum_{n=0}^{\infty} a_n \phi_n$$

converging uniformly to  $f$  on  $[-1, 1]$ .

**Definition 4. Schauder basis.** Let  $X$  be a Banach space. A sequence  $(e_n) \subset X$  is a *Schauder basis* if for every  $x \in X$  there exist unique scalars  $(c_n)$  such that the partial sums  $S_N = \sum_{n=0}^N c_n e_n$  converge to  $x$  in  $\|\cdot\|_X$ .

**Theorem 6. Faber–Schauder basis on  $[0, 1]$ .** Define  $\varphi_0(t) \equiv 1$ ,  $\varphi_1(t) = t$ , and for each  $m \geq 0$  and odd  $k \in \{1, 3, \dots, 2^m - 1\}$  set

$$\varphi_{m,k}(t) := \max\left(1 - 2^{m+1}\left|t - \frac{k}{2^m}\right|, 0\right).$$

Enumerate the system as  $\varphi_0, \varphi_1, \{\varphi_{m,k}\}_{m \geq 0, k \text{ odd}}$ . Then  $(\varphi_n)$  is a Schauder basis of  $C([0, 1])$  with respect to  $\|\cdot\|_{\infty}$ .

*Proof. Existence (uniform convergence).* For  $f \in C([0, 1])$  define coefficients

$$a_0 := f(0), \quad a_1 := f(1) - f(0), \quad a_{m,k} := f\left(\frac{k}{2^m}\right) - \frac{1}{2}\left(f\left(\frac{k-1}{2^m}\right) + f\left(\frac{k+1}{2^m}\right)\right).$$

Let  $S_M(f)$  be the finite linear combination of  $\varphi_0, \varphi_1$  and all  $\varphi_{m,k}$  with levels  $\leq M$ . A direct check shows  $S_M(f)$  is the polygonal interpolant of  $f$  on the dyadic grid  $\{j/2^M\}_{j=0}^{2^M}$ , hence (for the modulus of continuity  $\omega_f$ ) one has

$$\|f - S_M(f)\|_{\infty} \leq \omega_f(2^{-M}) \xrightarrow{M \rightarrow \infty} 0.$$

*Uniqueness.* Suppose  $\sum c_n \varphi_n \equiv 0$ . Evaluating at  $t = 0$  and  $t = 1$  gives  $c_0 = c_1 = 0$  (since all tents vanish at the endpoints). Next evaluate at the midpoint  $t = \frac{1}{2}$ : all functions from levels  $\geq 1$  vanish there, so  $c_{0,1} = 0$ . Inductively, at a new dyadic node  $t = \frac{k}{2^m}$  (odd  $k$ ) all higher-level tents vanish and lower levels are zero there; hence  $c_{m,k} = 0$ . Thus all coefficients vanish and the representation is unique.  $\square$

**Corollary 7. Transfer to  $[-1, 1]$ .** Let  $t = \frac{x+1}{2}$  be the affine bijection  $[-1, 1] \rightarrow [0, 1]$ . Define

$$\phi_0(x) := 1, \quad \phi_1(x) := \frac{x+1}{2}, \quad \phi_{m,k}(x) := \varphi_{m,k}\left(\frac{x+1}{2}\right).$$

Then  $(\phi_n) \subset C([-1, 1])$  is a Schauder basis. For every  $f \in C([-1, 1])$  there are unique coefficients  $(a_n)$  such that  $\sum_{n=0}^{\infty} a_n \phi_n$  converges uniformly to  $f$ .

**Remark 5. Explicit coefficient formulas on  $[-1, 1]$ .** Equivalently, for  $f \in C([-1, 1])$  let  $g(t) := f(2t - 1)$  on  $[0, 1]$ . Then

$$a_0 = g(0) = f(-1), \quad a_1 = g(1) - g(0) = f(1) - f(-1),$$

and for  $m \geq 0$ , odd  $k$ ,

$$a_{m,k} = g\left(\frac{k}{2^m}\right) - \frac{1}{2}\left(g\left(\frac{k-1}{2^m}\right) + g\left(\frac{k+1}{2^m}\right)\right) = f(x_{m,k}) - \frac{1}{2}(f(x_{m,k}^-) + f(x_{m,k}^+)),$$

where  $x_{m,k} = 2 \cdot \frac{k}{2^m} - 1$  and  $x_{m,k}^{\pm} = 2 \cdot \frac{k \pm 1}{2^m} - 1$ .



*Solution.* Apply the theorem on  $[0, 1]$  to  $g(t) := f(2t - 1)$  and pull back the expansion by  $t = \frac{x+1}{2}$ . Uniform convergence and uniqueness are preserved by this affine change of variables. Therefore the sequence  $(\phi_n)$  is as required.

**Example 1. Two quick checks.**

1. For  $f(t) = t$  on  $[0, 1]$ ,  $a_0 = 0$ ,  $a_1 = 1$ , and all  $a_{m,k} = 0$ ; hence  $f = \varphi_1$  exactly.
2. For  $f(t) = t^2$  on  $[0, 1]$ ,  $a_0 = 0$ ,  $a_1 = 1$ , and  $a_{m,k} = \frac{k^2}{4^m} - \frac{1}{2} \frac{(k-1)^2 + (k+1)^2}{4^m} = -4^{-m}$  (independent of  $k$ ), so  $t^2 = \varphi_1 - \sum_{m \geq 0} \sum_{k \text{ odd}} 4^{-m} \varphi_{m,k}$  with uniform convergence. Transfer to  $[-1, 1]$  via  $t = \frac{x+1}{2}$ .

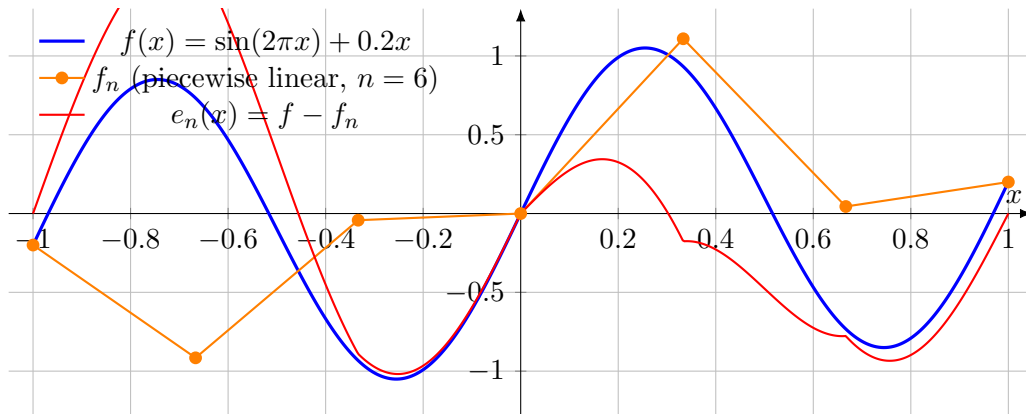


Figure 18: Function, piecewise-linear interpolant ( $n = 6$ ), and error curve.

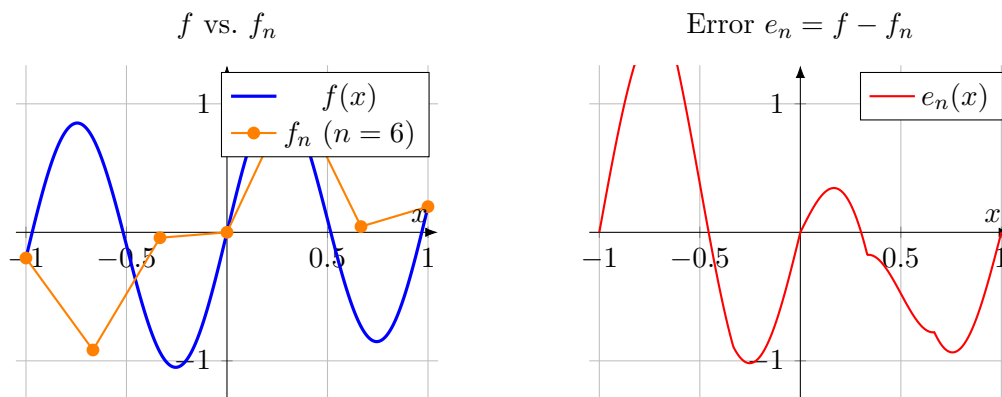


Figure 19: Left: function  $f$  and interpolant  $f_n$  ( $n = 6$ ). Right: error curve  $e_n = f - f_n$ .

**Exercise 6.** Powers preserve uniform convergence (under a bound). Let  $f_n : [0, 1] \rightarrow \mathbb{R}$  converge uniformly to  $f$ . Suppose  $f$  is bounded. Show that for every positive integer  $m$ ,

$$g_n(t) := f_n(t)^m \quad \text{converges uniformly on } [0, 1] \quad \text{to} \quad g(t) := f(t)^m.$$

**Definition 7. Uniform convergence.** We say  $f_n \rightarrow f$  uniformly on a set  $E$  if  $\sup_{t \in E} |f_n(t) - f(t)| \rightarrow 0$  as  $n \rightarrow \infty$ .

**Lemma 8. Power is Lipschitz on bounded intervals.** Fix  $m \in \mathbb{N}$  and  $M > 0$ . For all  $u, v \in [-M, M]$ ,

$$|u^m - v^m| \leq m M^{m-1} |u - v|.$$

*Proof.* By the Mean Value Theorem applied to  $\phi(x) = x^m$  on  $[-M, M]$ ,  $|u^m - v^m| = |\phi'(c)| |u - v| = m|c|^{m-1}|u - v| \leq mM^{m-1}|u - v|$  for some  $c$  between  $u$  and  $v$ . Alternatively, use  $u^m - v^m = (u - v) \sum_{k=0}^{m-1} u^{m-1-k} v^k$  and the bound  $|u|, |v| \leq M$ .  $\square$

**Theorem 8. Uniform convergence is preserved by powers.** Let  $f_n \rightarrow f$  uniformly on  $[0, 1]$  and suppose  $f$  is bounded. Then, for every  $m \in \mathbb{N}$ ,  $f_n^m \rightarrow f^m$  uniformly on  $[0, 1]$ .

*Proof.* Let  $B := \sup_{t \in [0, 1]} |f(t)| < \infty$ . Since  $f_n \rightarrow f$  uniformly, there exists  $N_0$  such that  $\sup_t |f_n(t) - f(t)| \leq 1$  for all  $n \geq N_0$ . Hence for  $n \geq N_0$ ,  $|f_n(t)| \leq B + 1 =: M$  for all  $t$ .

Fix  $m \in \mathbb{N}$ . By the lemma, for  $n \geq N_0$  and all  $t$ ,

$$|f_n(t)^m - f(t)^m| \leq m M^{m-1} |f_n(t) - f(t)|.$$

Taking suprema over  $t \in [0, 1]$ ,

$$\sup_t |f_n(t)^m - f(t)^m| \leq m M^{m-1} \sup_t |f_n(t) - f(t)| \xrightarrow{n \rightarrow \infty} 0,$$

so  $f_n^m \rightarrow f^m$  uniformly.  $\square$

**Example 2.** Let  $f(t) = \sin(2\pi t) + 0.2t$  and  $f_n(t) = f(t) + \frac{1}{n} \cos(6\pi t)$  on  $[0, 1]$ . Then  $\sup |f_n - f| \leq 1/n$ , so  $f_n \rightarrow f$  uniformly and  $f$  is bounded with  $\sup |f| \leq 1.2$ . Taking  $M = 2.2$ , for  $m = 3$ ,

$$\sup |f_n^3 - f^3| \leq 3M^2 \sup |f_n - f| \leq \frac{14.52}{n} \rightarrow 0.$$

Thus  $f_n^3 \rightarrow f^3$  uniformly; the same argument works for any  $m$ .

**Remark 9.** The boundedness of  $f$  is essential to produce a uniform Lipschitz constant for  $x \mapsto x^m$  on the range of  $f_n$  and  $f$ . More generally, if  $f_n \rightarrow f$  uniformly and  $\phi$  is uniformly continuous on a bounded interval containing the ranges of  $f$  and all large  $n$  of  $f_n$ , then  $\phi \circ f_n \rightarrow \phi \circ f$  uniformly.

**Exercise 10.** Runge approximation on a punctured domain. Let  $B_r(z)$  be the open disc of radius  $r$  about  $z \in \mathbb{C}$ , and set  $U := B_2(1) \setminus B_1(0)$ . Suppose  $f$  is holomorphic on  $U$ .

- (i) Show there exists a sequence converging to  $f$  uniformly on compact subsets of  $U$ .
- (ii) Must there be a sequence of polynomials converging to  $f$  uniformly on  $U$ ?
- (iii) If, moreover,  $f$  is holomorphic on an open set containing  $\overline{U}$ , must there be a sequence of polynomials converging to  $f$  uniformly on  $U$ ?

**Definition 11. Laurent polynomial.** A finite sum  $\sum_{k=-N}^M a_k z^k$  with integers  $N, M \geq 0$ .

**Theorem 9** (Runge, one-point-per-component form). If  $K \subset \mathbb{C}$  is compact and  $A \subset \widehat{\mathbb{C}} \setminus K$  contains at least one point from each component of  $\widehat{\mathbb{C}} \setminus K$ , then every function holomorphic on a neighborhood of  $K$  can be approximated uniformly on  $K$  by rational functions with poles only in  $A$ .

*Solution.* (i) Fix a compact  $K \subset U$ . Then  $\widehat{\mathbb{C}} \setminus K$  has two components: one contains  $\infty$ , the other contains  $0$ . Apply Runge with  $A = \{0, \infty\}$ . The approximants are rational functions with poles in  $A$ , hence Laurent polynomials. They converge uniformly to  $f$  on  $K$ . Since  $K$  was arbitrary, the convergence holds on every compact subset of  $U$ .

(ii) No. Take  $f(z) = 1/z$ . If polynomials  $p_n$  converged uniformly to  $f$  on  $U$ , then for any simple closed curve  $\gamma \subset U$  winding once around  $0$  we would have

$$\oint_{\gamma} p_n(z) dz \rightarrow \oint_{\gamma} \frac{1}{z} dz = 2\pi i,$$

but the left-hand side is 0 for all  $n$  (Cauchy's theorem), a contradiction.

(iii) Still no in general. The same  $f(z) = 1/z$  is holomorphic on  $\mathbb{C} \setminus \{0\}$ , which contains  $\overline{U}$ , yet cannot be uniformly approximated on  $U$  by polynomials, by the argument in (ii).  $\square$

*Remark 12.* While polynomials fail in (ii)–(iii), Runge guarantees uniform approximation on  $U$  by *rational* (indeed, Laurent-polynomial) functions with poles at 0 (and possibly at  $\infty$ ).

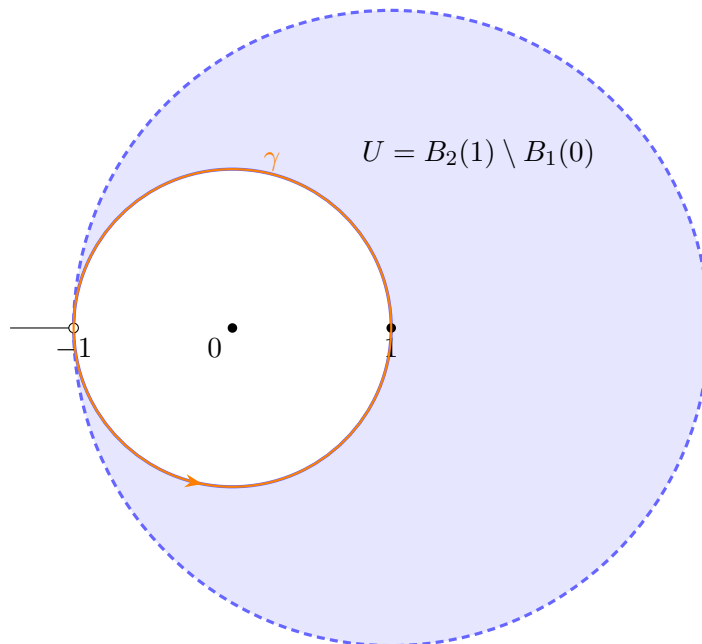


Figure 20: Domain  $U$ : open disc  $B_2(1)$  with the open unit disc  $B_1(0)$  removed. Dashed boundary is not in  $U$ ; the inner circle is.

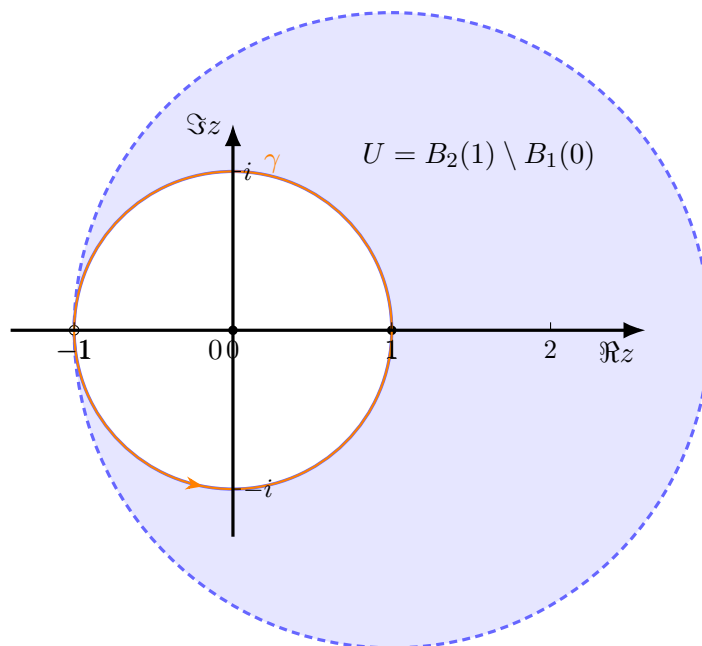


Figure 21: Domain  $U$  with axes and ticks. Dashed boundary is excluded; inner circle is included.

**Example 3.** *Approximating  $1/z$  on  $U = B_2(1) \setminus B_1(0)$ .*

(i) (Compacts in  $U$ .) Runge with poles at  $\{0, \infty\}$  allows Laurent polynomials. For  $f(z) = 1/z$  the constant sequence

$$r_n(z) \equiv \frac{1}{z}$$

already works and converges uniformly to  $1/z$  on every compact  $K \subset U$ .

If  $K \subset B_1(1)$  (so  $|z - 1| < 1$  on  $K$ ), the polynomials

$$p_N(z) = \sum_{k=0}^N (-1)^k (z - 1)^k$$

satisfy  $p_N \rightarrow 1/z$  uniformly on  $K$  by the geometric series  $\frac{1}{1+w} = \sum_{k \geq 0} (-w)^k$  with  $w = z - 1$ .

(ii) (No polynomials on all of  $U$ .) Suppose  $p_n$  were polynomials with  $p_n \rightarrow 1/z$  uniformly on  $U$ . For the loop  $\gamma(t) = e^{it}$ ,

$$\int_{\gamma} p_n(z) dz \longrightarrow \int_{\gamma} \frac{1}{z} dz = 2\pi i,$$

but each  $\int_{\gamma} p_n(z) dz = 0$ , contradiction. Hence no polynomial sequence converges to  $1/z$  uniformly on  $U$  (and the same holds even if  $1/z$  extends past  $\overline{U}$ ).

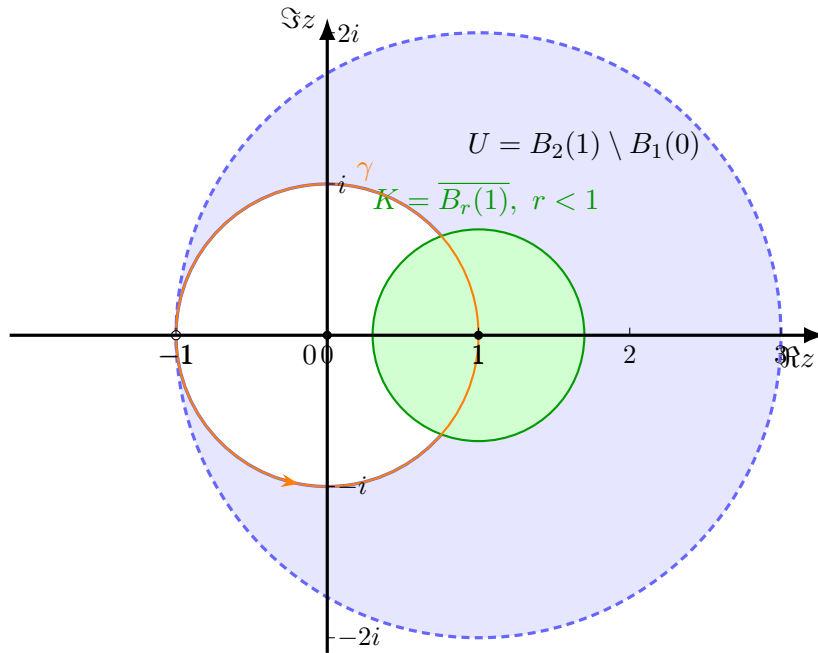


Figure 22: Domain  $U$  (blue), a compact  $K \subset B_1(1)$  (green) where polynomials approximate  $1/z$ , and a loop  $\gamma$  around  $0$  used to rule out uniform polynomial approximation on all of  $U$ . Axes are extended for clarity.

**Definition 13. Runge set / pole set.** For a compact  $K \subset \mathbb{C}$  and a set  $A \subset \widehat{\mathbb{C}} \setminus K$ , we say  $A$  is a *Runge pole set* for  $K$  if  $A$  meets every connected component of  $\widehat{\mathbb{C}} \setminus K$ .

**Theorem 10. Runge's theorem (rational form).** Let  $K \subset \mathbb{C}$  be compact and let  $A \subset \widehat{\mathbb{C}} \setminus K$  meet every component of  $\widehat{\mathbb{C}} \setminus K$ . If  $f$  is holomorphic on a neighbourhood of  $K$ , then there exists a sequence of rational functions  $\{r_n\}$  with all poles contained in  $A$  such that

$$\sup_{z \in K} |r_n(z) - f(z)| \xrightarrow{n \rightarrow \infty} 0.$$

**Corollary 11. Polynomial approximation (connected complement).** If  $\widehat{\mathbb{C}} \setminus K$  is connected, then one may take  $A = \{\infty\}$ , hence  $f$  can be uniformly approximated on  $K$  by polynomials.

**Corollary 12. Laurent–polynomial approximation (one bounded hole).** If  $\widehat{\mathbb{C}} \setminus K$  has exactly two components (the unbounded one and a bounded one containing a point  $a$ ), then taking  $A = \{\infty, a\}$  shows that  $f$  can be uniformly approximated on  $K$  by finite Laurent series in powers of  $(z - a)$  (i.e. by sums  $\sum_{k=-N}^M a_k(z - a)^k$ ).

**Remark 14. Specialization to  $U = B_2(1) \setminus B_1(0)$ .** For any compact  $K \subset U$ , the complement  $\widehat{\mathbb{C}} \setminus K$  has two components: one containing  $\infty$ , one containing 0. Runge with  $A = \{\infty, 0\}$  yields uniform approximation on  $K$  by Laurent polynomials with possible pole at 0. In particular, for  $f(z) = 1/z$ , the constant sequence  $r_n(z) \equiv 1/z$  (a rational function with pole at 0) already converges uniformly on every compact  $K \subset U$ . By contrast, no sequence of *polynomials* can converge uniformly to  $1/z$  on all of  $U$  (use a loop  $\gamma \subset U$  winding once around 0 and  $\int_\gamma p(z) dz = 0$  versus  $\int_\gamma \frac{1}{z} dz = 2\pi i$ ).

**Exercise 15.** Polynomial approximation of  $1/z$  on a semicirclepoly-1-over-z-semi Construct a sequence of polynomials that converges uniformly to  $1/z$  on

$$K = \{z \in \mathbb{C} : |z| = 1, \Re z \geq 0\}.$$

**Definition 16. Runge/Mergelyan (polynomial case).** If  $K \subset \mathbb{C}$  is compact with connected complement, every function continuous on  $K$  and holomorphic on  $\text{int } K$  can be uniformly approximated on  $K$  by polynomials.

*Construction and proof of convergence.* Set  $t_k^{(n)} = -\frac{\pi}{2} + \frac{k\pi}{n}$  and  $z_k^{(n)} = e^{it_k^{(n)}}$  for  $k = 0, \dots, n$ . Let

$$\ell_k^{(n)}(z) = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{z - z_j^{(n)}}{z_k^{(n)} - z_j^{(n)}}, \quad p_n(z) = \sum_{k=0}^n \frac{1}{z_k^{(n)}} \ell_k^{(n)}(z).$$

Then  $p_n$  is a polynomial and  $p_n(z_k^{(n)}) = 1/z_k^{(n)}$  for all  $k$ . Since  $K$  is an arc (its complement is connected) and  $1/z$  is analytic on a neighbourhood of  $K$ , standard Bernstein–Walsh estimates for analytic data on Jordan arcs imply  $p_n \rightarrow 1/z$  uniformly on  $K$  (in fact,  $\|p_n - 1/z\|_K \leq C\rho^{-n}$  for some  $\rho > 1$ ). This provides the required sequence.  $\square$

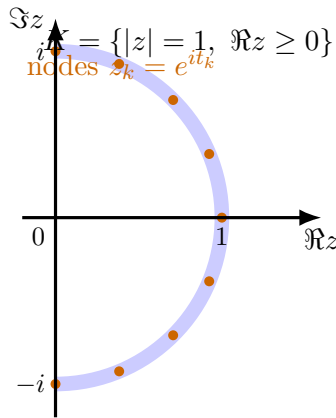
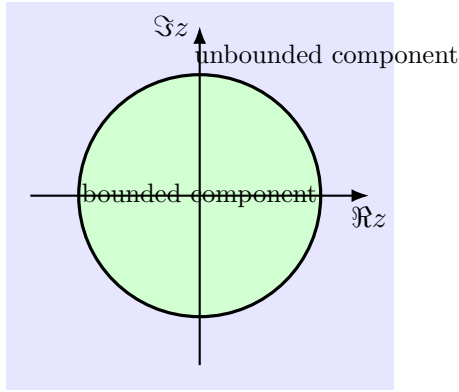
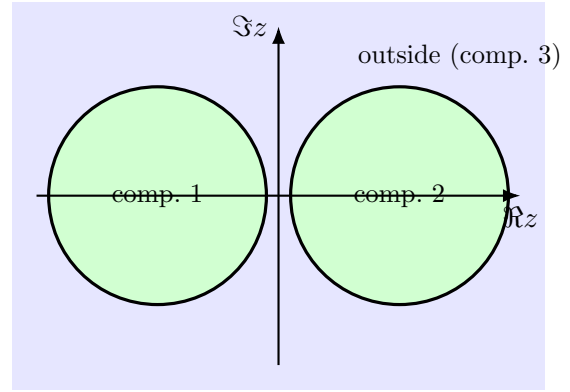


Figure 23: Right semicircle  $K$  and equally spaced interpolation nodes for the Lagrange polynomials  $p_n$ .



$K = \{|z| = 1\}$ ,  $\mathbb{C} \setminus K$  has 2 components



$K = C_1 \cup C_2$ ,  $\mathbb{C} \setminus K$  has 3 components

Figure 24: Complements of compact sets can be disconnected: a single circle separates the plane into two components; two disjoint circles give three components.

**Example 4** (What the dots mean; explicit  $p_2$ ). Let  $K = \{|z| = 1, \Re z \geq 0\}$  and choose nodes  $z_0 = -i$ ,  $z_1 = 1$ ,  $z_2 = i$ . The Lagrange interpolant to  $f(z) = 1/z$  is

$$p_2(z) = \sum_{k=0}^2 \frac{1}{z_k} \prod_{j \neq k} \frac{z - z_j}{z_k - z_j} = z^2 - z + 1.$$

Hence  $p_2(1) = 1$ ,  $p_2(i) = -i$ ,  $p_2(-i) = i$ , i.e.  $p_2$  matches  $1/z$  at the three orange nodes on the arc. For general  $n$ , pick  $n+1$  nodes on  $K$ , form  $p_n$  by the same formula; then  $p_n \rightarrow 1/z$  uniformly on  $K$ .

**Exercise 17.** Construct explicit polynomials  $p_n \rightarrow 1/z$  on the right semicircle. Let

$$K := \{z \in \mathbb{C} : |z| = 1, \Re z \geq 0\}.$$

Construct polynomials  $p_n$  such that  $p_n \rightarrow 1/z$  uniformly on  $K$ .

**Definition 18** (Lagrange interpolation). Given  $n+1$  distinct nodes  $z_0, \dots, z_n \in \mathbb{C}$ , the Lagrange basis polynomials are

$$\ell_k(z) := \prod_{\substack{j=0 \\ j \neq k}}^n \frac{z - z_j}{z_k - z_j}, \quad k = 0, \dots, n,$$

so that  $\ell_k(z_j) = \delta_{kj}$  and  $\sum_{k=0}^n \ell_k \equiv 1$ . For a function  $f$  defined on the nodes, the (unique) degree- $\leq n$  interpolant is

$$I_n[f](z) := \sum_{k=0}^n f(z_k) \ell_k(z).$$

*Remark 19* (Barycentric evaluation (stable)). With weights  $w_k := (\prod_{j \neq k} (z_k - z_j))^{-1}$ ,

$$I_n[f](z) = \frac{\sum_{k=0}^n \frac{w_k f(z_k)}{z - z_k}}{\sum_{k=0}^n \frac{w_k}{z - z_k}},$$

which is numerically stable and avoids forming  $\ell_k$  explicitly.

**Definition 20** (Our nodes on the arc). For each  $n \in \mathbb{N}$ , choose equally spaced angles

$$t_k^{(n)} := -\frac{\pi}{2} + \frac{k\pi}{n}, \quad k = 0, \dots, n,$$

and nodes on the right semicircle

$$z_k^{(n)} := e^{it_k^{(n)}} \in K.$$

(Other admissible choices are *Chebyshev-distributed* or *Leja* nodes on  $K$ .)

**Theorem 13** (Construction). Let  $f(z) = 1/z$ . For  $n \geq 1$  define the polynomial

$$p_n(z) := \sum_{k=0}^n \frac{1}{z_k^{(n)}} \ell_k^{(n)}(z), \quad \ell_k^{(n)}(z) := \prod_{\substack{j=0 \\ j \neq k}}^n \frac{z - z_j^{(n)}}{z_k^{(n)} - z_j^{(n)}}.$$

Then  $p_n$  is a polynomial of degree  $\leq n$  and  $p_n(z_k^{(n)}) = 1/z_k^{(n)}$  for all  $k$ .

**Theorem 14** (Uniform convergence on  $K$ ). With the above nodes,  $p_n \rightarrow 1/z$  uniformly on  $K$ .

Why the convergence holds (standard argument). The set  $K$  is a compact Jordan arc; its complement is connected. The function  $1/z$  is holomorphic on a fixed annulus  $\{1 - \delta < |z| < 1 + \delta\} \supset K$ . By the *Bernstein–Walsh estimate*, the best approximation error

$$E_n := \inf_{\deg q \leq n} \|1/z - q\|_K$$

decays geometrically:  $E_n \leq C\rho^{-n}$  for some  $\rho > 1$ . For the Lagrange interpolant,

$$\|1/z - p_n\|_K \leq (1 + \Lambda_n) E_n, \quad \Lambda_n := \sup_{z \in K} \sum_{k=0}^n |\ell_k^{(n)}(z)| \quad (\text{Lebesgue constant}).$$

For reasonable node families on smooth/analytic arcs (e.g. Chebyshev-type or Leja nodes),  $\Lambda_n$  grows at most polynomially in  $n$ , hence  $(1 + \Lambda_n)E_n \rightarrow 0$  geometrically. Therefore  $p_n \rightarrow 1/z$  uniformly on  $K$ .  $\square$

**Example 5** (Make  $p_n$  concrete:  $n = 2$ ). Take three nodes  $z_0 = -i$ ,  $z_1 = 1$ ,  $z_2 = i$  on  $K$  (angles  $-\frac{\pi}{2}, 0, \frac{\pi}{2}$ ). Then

$$p_2(z) = \sum_{k=0}^2 \frac{1}{z_k} \ell_k(z) = i \ell_0(z) + 1 \cdot \ell_1(z) - i \ell_2(z) \quad \Rightarrow \quad \boxed{p_2(z) = z^2 - z + 1}.$$

Indeed  $p_2(1) = 1$ ,  $p_2(i) = -i$ ,  $p_2(-i) = i$  (matching  $1/z$  at the nodes). On  $K$ ,  $|p_2(e^{it}) - e^{-it}| = \sqrt{(\cos 2t + 1 - 2 \cos t)^2 + (\sin 2t)^2}$ , so the uniform error already is modest and will decay rapidly as  $n$  grows.

**Remark 21** (Geometric meaning). On the unit circle,  $1/z = \bar{z}$ . Thus we are approximating the *conjugation* map restricted to the right semicircle by polynomials in  $z$ . This is possible uniformly on the arc because its complement is connected (Mergelyan/Runge), even though conjugation is anti-holomorphic globally.

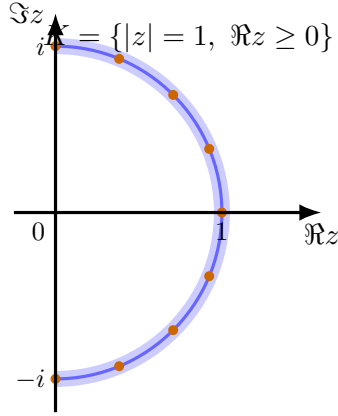


Figure 25: Right semicircle  $K$  and equally spaced interpolation nodes  $z_k = e^{it_k}$ .

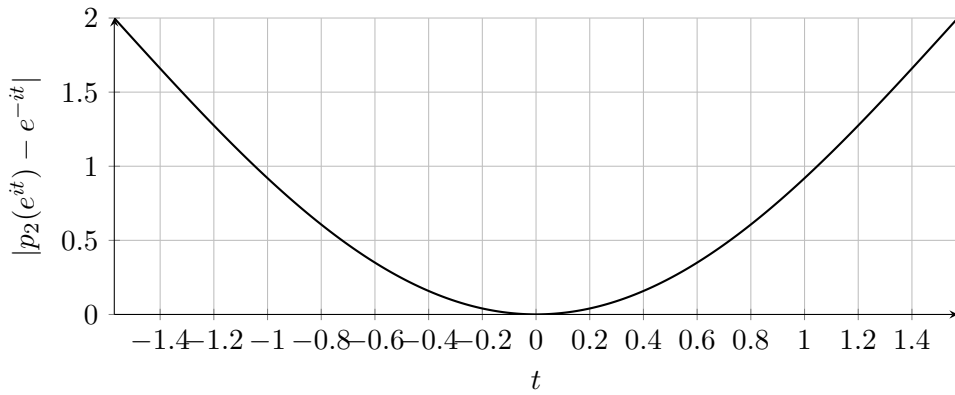


Figure 26: Pointwise error of  $p_2(z) = z^2 - z + 1$  along  $z = e^{it}$ ,  $t \in [-\pi/2, \pi/2]$ .

**Exercise 22.** Polynomial characterization of holomorphicity on Runge domainspoly-char Let  $U \subset \mathbb{C}$  be bounded, open, and suppose  $\mathbb{C} \setminus U$  is connected. Show that  $f : U \rightarrow \mathbb{C}$  is holomorphic iff for every compact  $K \Subset U$  and every  $\varepsilon > 0$  there exists a polynomial  $P$  with

$$\sup_{z \in K} |f(z) - P(z)| < \varepsilon.$$

**Definition 23. Polynomial hull in the plane.** For a compact  $K \subset \mathbb{C}$ , let

$$\widehat{K} := K \cup \bigcup_{\Omega \text{ bounded component of } \mathbb{C} \setminus K} \overline{\Omega}.$$

Then  $\mathbb{C} \setminus \widehat{K}$  is connected. If  $K \Subset U$  and  $\mathbb{C} \setminus U$  is connected, then  $\widehat{K} \Subset U$ .

**Theorem 15** (Mergelyan). *If  $K \subset \mathbb{C}$  is compact with connected complement, and  $g$  is continuous on  $K$  and holomorphic on  $\text{int } K$ , then for every  $\varepsilon > 0$  there exists a polynomial  $P$  with  $\sup_K |g - P| < \varepsilon$ .*

*Solution.* ( $\Rightarrow$ ) Assume  $f$  holomorphic on  $U$ , let  $K \Subset U$ , and set  $\widehat{K}$  as above. Then  $\widehat{K} \Subset U$  and  $\mathbb{C} \setminus \widehat{K}$  is connected, hence by Mergelyan there is a polynomial  $P$  with  $\sup_{\widehat{K}} |f - P| < \varepsilon$ , thus in particular on  $K$ .

( $\Leftarrow$ ) Assume: for every  $K \Subset U$  and  $\varepsilon > 0$  there is a polynomial  $P$  with  $\sup_K |f - P| < \varepsilon$ . Let  $K_1 \Subset K_2 \Subset \dots \Subset U$  with  $\bigcup_m K_m = U$  and choose polynomials  $P_m$  with  $\sup_{K_m} |f - P_m| < 1/m$ .



Then  $P_m \rightarrow f$  uniformly on each  $K_\ell$ , so by Weierstrass' theorem the limit  $f$  is holomorphic on  $U$ .  $\square$

*Remark 24.* On  $U = \{ |z| < 1 \}$  one may take  $P_n$  to be partial sums of the Taylor series of  $f$  at 0, which converge uniformly on  $K = \{ |z| \leq r \}$  for each  $r < 1$ . The theorem shows such polynomial approximation on every compact holds on any bounded  $U$  with connected complement.

**Definition 25. Compact inclusion.** For sets  $K \subset U \subset \mathbb{C}$ , we write  $K \Subset U$  if  $K \subset U$  and  $\overline{K}$  is compact with  $\overline{K} \subset U$ . Equivalently, in the plane  $\text{dist}(K, \partial U) > 0$ .

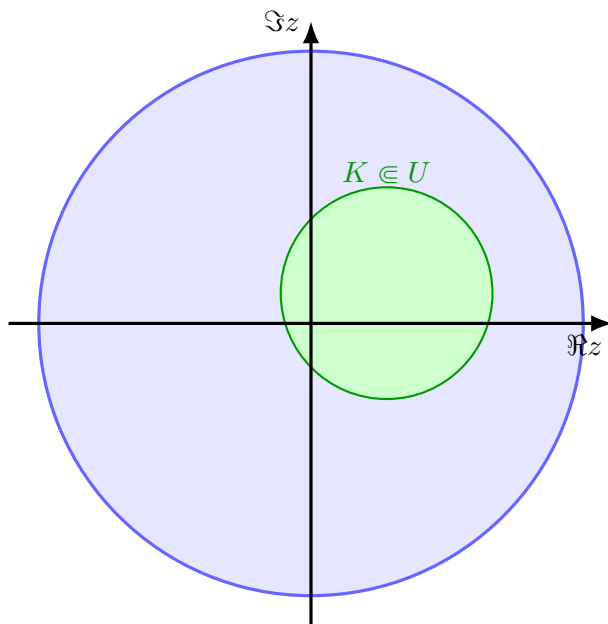


Figure 27:  $K \Subset U$ :  $K$  sits a positive distance away from  $\partial U$ .

**Definition 26. Pointwise and uniform convergence.** Let  $E$  be a set and  $(f_n)$  functions on  $E$ .

- $f_n \rightarrow f$  pointwise on  $E$  if  $\forall x \in E, \forall \varepsilon > 0, \exists N = N(\varepsilon, x)$  with  $|f_n(x) - f(x)| < \varepsilon$  for  $n \geq N$ .
- $f_n \rightarrow f$  uniformly on  $E$  if  $\sup_{x \in E} |f_n(x) - f(x)| \xrightarrow{n \rightarrow \infty} 0$ .
- $f_n \rightarrow f$  locally uniformly on an open  $U$  if for every  $K \Subset U$ , the convergence is uniform on  $K$ .

*Remark 27.* On a compact set  $K$ , uniform on  $K \iff$  convergence in the sup norm  $\| \cdot \|_\infty$ . If  $U$  is open, “locally uniform” means uniform on every  $K \Subset U$ ; on a compact  $U$  this reduces to uniform on  $U$ .

**Example 6** (Uniform vs nonuniform convergence).

1. Uniform:  $f_n(x) = \frac{x}{n+x}$  on  $[0, 1]$ ; then  $\sup_{[0,1]} |f_n| = \frac{1}{n+1} \rightarrow 0$ , so  $f_n \rightarrow 0$  uniformly.
2. Nonuniform:  $f_n(x) = x^n$  on  $[0, 1]$ ; then  $f_n \rightarrow f$  pointwise with  $f = \mathbf{1}_{\{1\}}$ , but not uniformly, since  $\sup_{x \in [0,1]} |x^n - 0| = 1$  for all  $n$ .

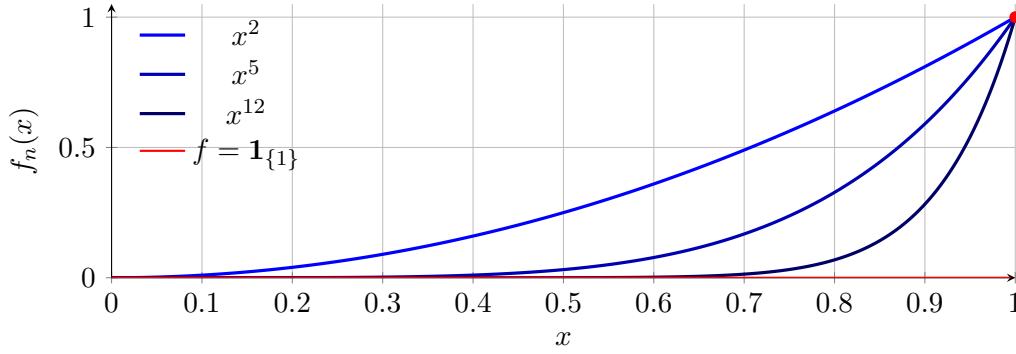


Figure 28:  $x^n \rightarrow \mathbf{1}_{\{1\}}$  pointwise on  $[0, 1]$ , but not uniformly.

**Example 7** (Uniform vs nonuniform approximation by series). Let  $S_n(x) = \sum_{k=0}^n x^k$ .

- On  $[0, b]$  with  $0 < b < 1$ :  $\left\| \frac{1}{1-x} - S_n \right\|_{[0,b]} = \sup_{[0,b]} \frac{x^{n+1}}{1-x} \leq \frac{b^{n+1}}{1-b} \rightarrow 0$  (uniform).
- On  $[0, 1]$ : not uniform, since  $\sup_{x \in [0,1]} \frac{x^{n+1}}{1-x} = +\infty$ .

**Exercise 28.** Polynomials  $p_n$  with  $p_n(0) = 1$  and  $p_n(z) \rightarrow 0$  for  $z \neq 0$  poly-vanish-off-zero Does there exist a sequence of complex polynomials  $(p_n)$  such that  $p_n(0) = 1$  for all  $n$  and  $p_n(z) \rightarrow 0$  for each  $z \in \mathbb{C} \setminus \{0\}$ ?

**Definition 29.** For  $n \in \mathbb{N}$  let

$$K_n := \left\{ z : \frac{1}{n} \leq |z| \leq n, \quad |\arg z| \leq \pi - \frac{1}{n} \right\} \quad (\text{a “slit annulus”}).$$

Then  $K_n$  is compact,  $K_n \subseteq \mathbb{C} \setminus \{0\}$ , and  $\bigcup_{n \geq 1} K_n = \mathbb{C} \setminus \{0\}$ . The complement  $\mathbb{C} \setminus K_n$  is connected because the missing thin sector joins the inner and outer parts.

**Theorem 16.** There exists a sequence of polynomials  $(p_n)$  with  $p_n(0) = 1$  for all  $n$  and  $p_n(z) \rightarrow 0$  for every  $z \neq 0$ .

*Proof.* Fix  $n$ . Define a function on the compact set  $E_n := K_n \cup \{0\}$  by

$$\phi_n(z) = \begin{cases} 0, & z \in K_n, \\ 1, & z = 0. \end{cases}$$

This  $\phi_n$  is continuous on  $E_n$  and holomorphic on  $\text{int } E_n$  (the interior is the slit annulus, where  $\phi_n \equiv 0$ ). Since  $\mathbb{C} \setminus E_n = (\mathbb{C} \setminus K_n) \setminus \{0\}$  is still connected, Mergelyan’s theorem applies: for each  $\varepsilon_n > 0$  there exists a polynomial  $P_n$  with

$$\sup_{z \in E_n} |P_n(z) - \phi_n(z)| < \varepsilon_n.$$

Choose  $\varepsilon_n \leq \frac{1}{4}$  so that in particular  $|P_n(0) - 1| < \frac{1}{4}$ . Set

$$p_n(z) := \frac{P_n(z)}{P_n(0)}.$$

Then  $p_n$  is a polynomial with  $p_n(0) = 1$ , and for  $z \in K_n$ ,

$$|p_n(z)| = \frac{|P_n(z)|}{|P_n(0)|} \leq \frac{|P_n(z) - \phi_n(z)|}{|P_n(0)|} \leq \frac{\varepsilon_n}{1 - \varepsilon_n} \leq 2\varepsilon_n.$$

Now fix  $z \neq 0$ . For  $n$  large enough we have  $z \in K_n$ , hence  $|p_n(z)| \leq 2\varepsilon_n \rightarrow 0$  as  $n \rightarrow \infty$ . This proves the claim.  $\square$

*Remark 30.* No uniformity on large discs is possible: if a polynomial  $q$  were tiny on the circle  $|z| = R$ , the maximum modulus principle on  $|z| \leq R$  would force  $|q(0)|$  tiny as well. The slit ensures  $\mathbb{C} \setminus K_n$  is connected, allowing Mergelyan, but  $K_n$  does *not* enclose 0, so there is no contradiction with  $p_n(0) = 1$ .

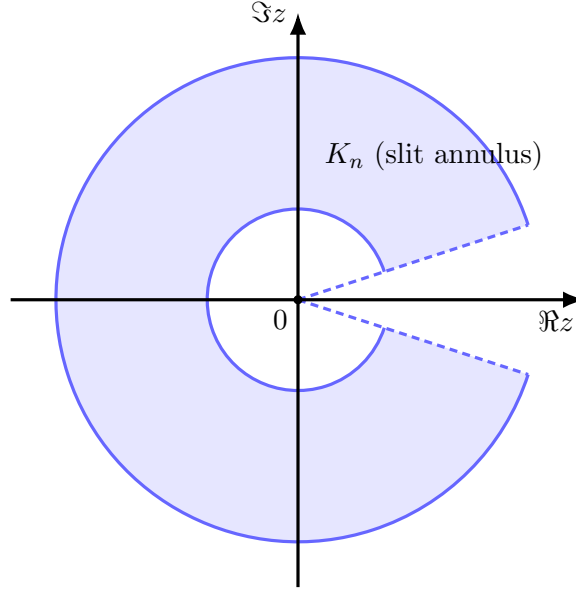


Figure 29: A typical  $K_n$ : annulus  $\{1/n \leq |z| \leq n\}$  minus a thin sector; its complement is connected.

**Exercise 31.** Uniform polynomial approximation on an annulus forces extension -annulus-extension Let  $A = \{z \in \mathbb{C} : \frac{1}{2} \leq |z| \leq 1\}$  and let  $f : A \rightarrow \mathbb{C}$  be continuous on  $A$  and holomorphic on  $\text{int } A$ . Suppose polynomials  $p_n$  converge uniformly to  $f$  on  $A$ . Show there exists  $g : \{|z| \leq 1\} \rightarrow \mathbb{C}$  continuous, holomorphic on  $\{|z| < 1\}$ , with  $g = f$  on  $A$ .

*Proof.* Fix  $\rho \in (\frac{1}{2}, 1)$ . Since  $|z| = \rho \subset A$  and  $p_n \rightarrow f$  uniformly on  $A$ , the sequence  $(p_n)$  is uniformly Cauchy on  $|z| = \rho$ :  $\sup_{|z|=\rho} |p_n - p_m| \rightarrow 0$  as  $n, m \rightarrow \infty$ . For each  $n, m$  the difference  $p_n - p_m$  is entire, hence by the maximum modulus principle on the closed disk  $\overline{D}_\rho := \{|z| \leq \rho\}$ ,

$$\sup_{|z| \leq \rho} |p_n(z) - p_m(z)| \leq \sup_{|z|=\rho} |p_n(z) - p_m(z)| \xrightarrow{n, m \rightarrow \infty} 0.$$

Thus  $(p_n)$  converges uniformly on  $\overline{D}_\rho$  to some function  $g_\rho$  that is holomorphic on  $|z| < \rho$ .

If  $\rho' < \rho$  are both  $> \frac{1}{2}$ , then  $p_n \rightarrow g_{\rho'}$  on  $\overline{D}_{\rho'}$  and  $p_n \rightarrow g_\rho$  on  $\overline{D}_\rho$ , so  $g_{\rho'} = g_\rho$  on  $|z| < \rho'$ . Define  $g$  on  $\{|z| < 1\} = \bigcup_{\rho > 1/2} \{|z| < \rho\}$  by  $g := g_\rho$  on  $|z| < \rho$ ; this is well defined and holomorphic.

On  $A \cap \{|z| \leq \rho\} = \{\frac{1}{2} \leq |z| \leq \rho\}$  we have  $p_n \rightarrow f$  (by hypothesis) and  $p_n \rightarrow g$  (by uniform convergence on  $\overline{D}_\rho$ ), hence  $g = f$  there. Letting  $\rho \uparrow 1$  yields  $g = f$  on all of  $A$ . Finally,  $g$  is continuous on  $\{|z| \leq 1\}$  because it is holomorphic on  $\{|z| < 1\}$  and coincides with the continuous  $f$  on  $|z| = 1$ .  $\square$

*Remark 32.* As a corollary,  $1/z$  cannot be approximated uniformly by polynomials on  $A$ , since such an approximation would produce a holomorphic extension across 0.

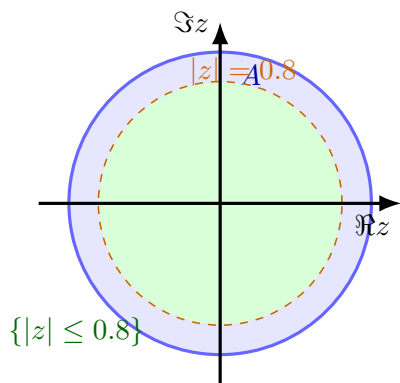


Figure 30: Fix  $\rho \in (\frac{1}{2}, 1)$ . Uniform convergence on  $|z| = \rho$  and MMP make  $(p_n)$  Cauchy on  $\{|z| \leq \rho\}$ , giving a holomorphic limit inside; letting  $\rho \uparrow 1$  fills the disk and matches  $f$  on  $A$ .

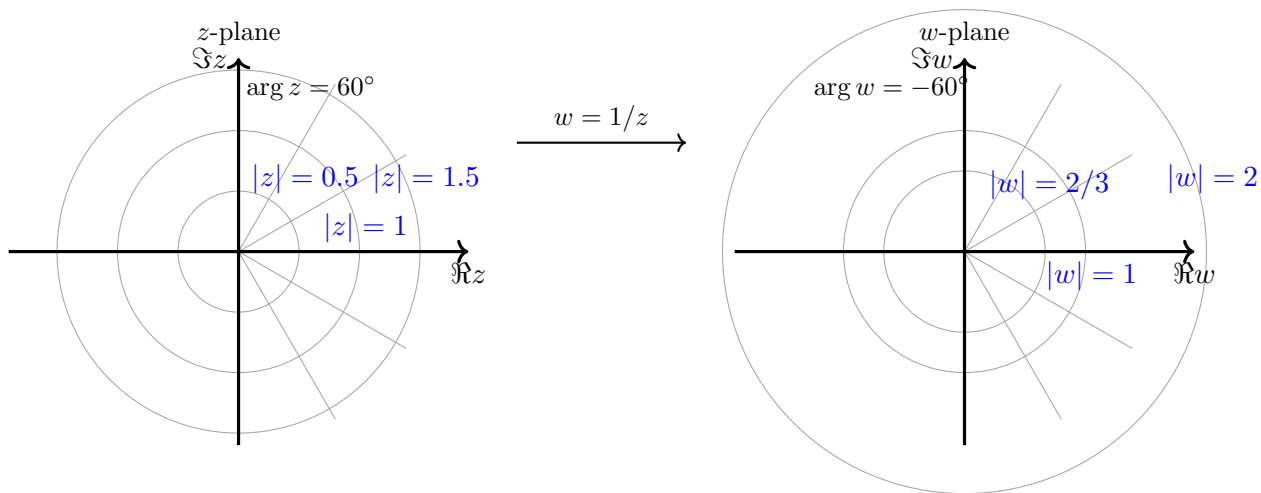


Figure 31: Circles centered at 0 invert radii, rays flip angles. The unit circle is fixed (parameters reverse).

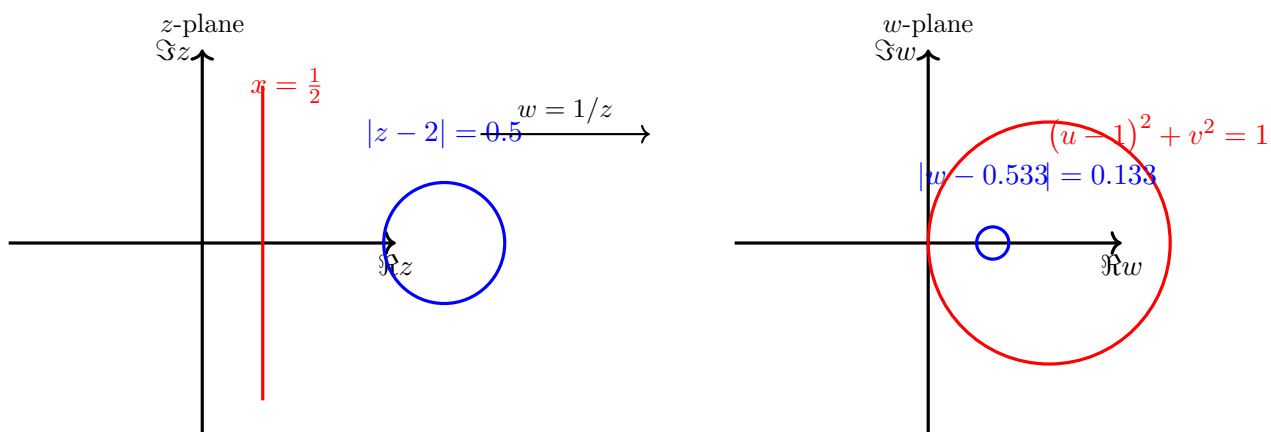


Figure 32: Under  $w = 1/z$ : lines not through 0 become circles through 0; circles not through 0 remain circles with transformed center/radius.